

INSTANT PAYMENT SWITCH (IPS)

Business Process Flows

IPSL

Introduction

Integrated Payment Services Limited (IPSL) is in the process of upgrading PESALINK platform from the current switch running on ISO8583 standard to a new Instant Payment Switch (IPS) running on ISO20022.

The major reason that informed IPSL to upgrade is that the existing SWITCH runs on ISO8583 as the financial industry is moving away from ISO8583 to ISO20022 message standard.

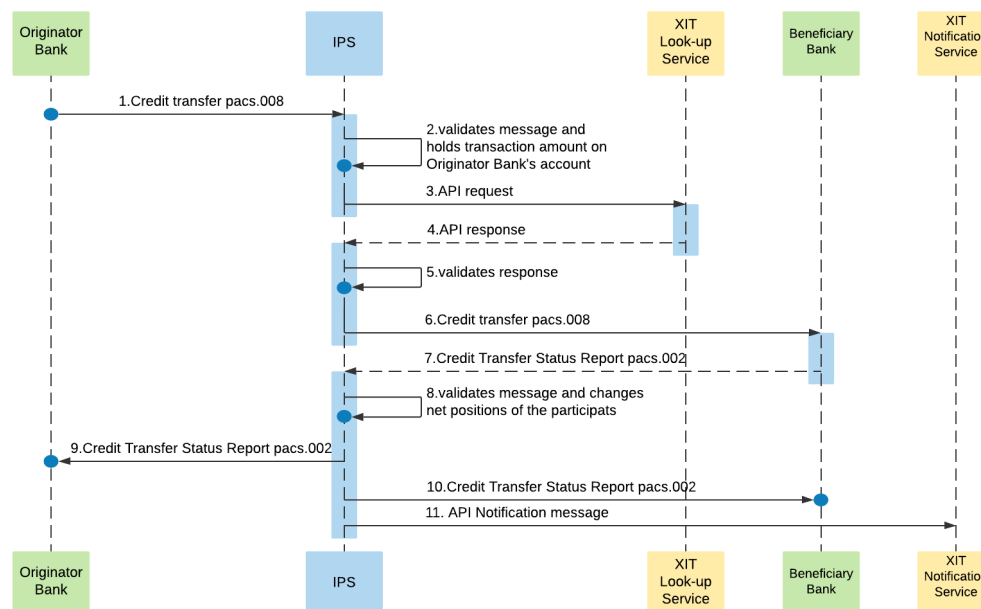
Below are the functionalities that will be provided by the new IPS:

- Account validation – This will be done against receiver bank's data via API.
- Support of different types of payments (Credit Transfers and Request to Pay, Request to Pay by PSP and Request to Pay with Mandate ID);
- Support the three-layer structure of participants (direct, indirect and payment initiators or sub participants);
- Limit management system for payment transaction processing;
- Fee management system;
- Reports
- Anti-Fraud checks

The IPS is designed to provide secure payment and information exchange via secure network.

Access to IPS services is provided for eligible Participants through Unified REST API.

Below are the different business flows:

1 Credit Transfer to phone number, positive scenario

1. The Originator Bank submits Credit Transfer message pacs.008 to the IPS. This message contains the recipients' name, surname, and phone number instead of the account number of the recipient.
2. The IPS validates the message, checks statuses of the Originator Bank and Beneficiary Bank, checks existing limits for the particular participants, and checks the IPS net position of the Originator Bank. The IPS makes a reservation for the Credit Transfer amount on the IPS account of the Originator Bank.
3. The IPS initiates an API call to XIT Look-up Service to request the account number associated with the provided phone number and the bank where the account is opened.
4. The XIT Look-up Service returns the account number associated with the phone number provided in the request message and the bank where the account is opened.
5. The IPS validated the received response message (whether received account number corresponds to the format standards).
6. The IPS replaces phone number with the corresponding account number received from the XIT Look-up Service and forwards the Credit Transfer message pacs.008 to the Beneficiary Bank.
7. The Beneficiary Bank receives, validates positively the Credit Transfer message pacs.008 and responses with the positive confirmation within the Credit Transfer Status Report message pacs.002.
8. The IPS positively validates the received response Credit Transfer Status Report message and changes net positions on the participants by moving Credit Transfer amount (including fee, if applicable) from the Originator Bank's IPS account to Beneficiary Bank's IPS account.
9. The IPS submits Credit Transfer Status Report message pacs.002 to the Originator Bank. Based on this message the Originator Bank may inform its customer (Ultimate Debtor) about the successful transaction.
10. The IPS submits Credit Transfer Status Report message pacs.002 to the Beneficiary Bank. Based on this message the Beneficiary Bank may finalize the transaction with its customer (Ultimate Creditor) by making the Credit Transfer amount available on the customers' account.

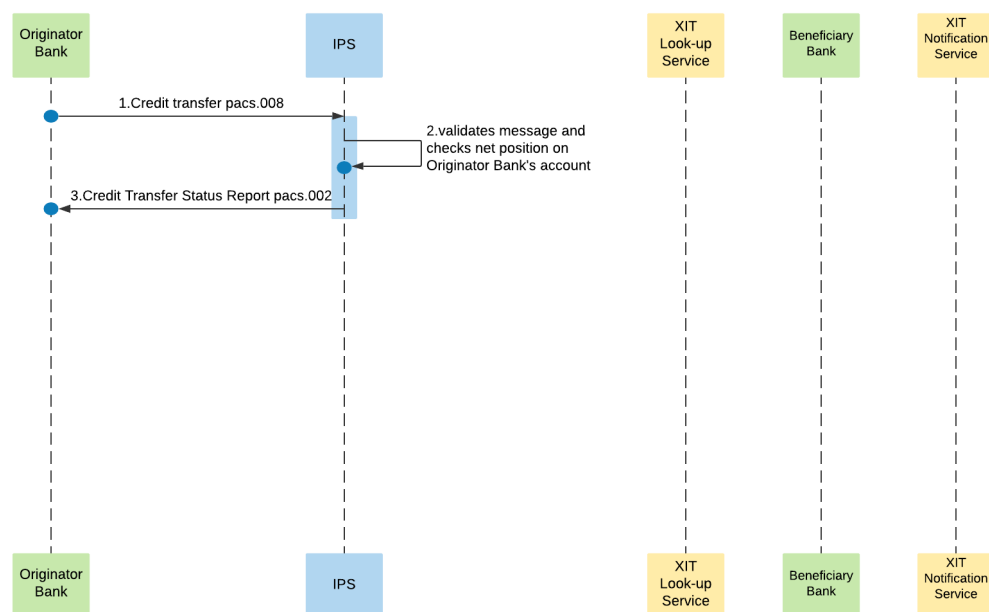
Note: Message exchange between the parties is controlled by the application layer under HTTPs.

If the IPS receives the response from the network that the message Credit Transfer Status Report (pacs.002) is not delivered to the Beneficiary Bank, the IPS is allowed additionally 4 (four) times to re-submit the Credit Transfer Status Report message to the Beneficiary Bank. If after these 4 attempts the Credit Transfer Status Report (pacs.002) still is not delivered to Beneficiary Bank, before finalizing the credit transaction with the Ultimate Creditor the Beneficiary Bank must initiate Investigation message (pacs.028).

11. Additionally, after successful transaction processing the IPS generate a notification API message for XIT Notification Service for further SMS delivery to the customer.

2 Credit Transfer to phone number, negative scenarios

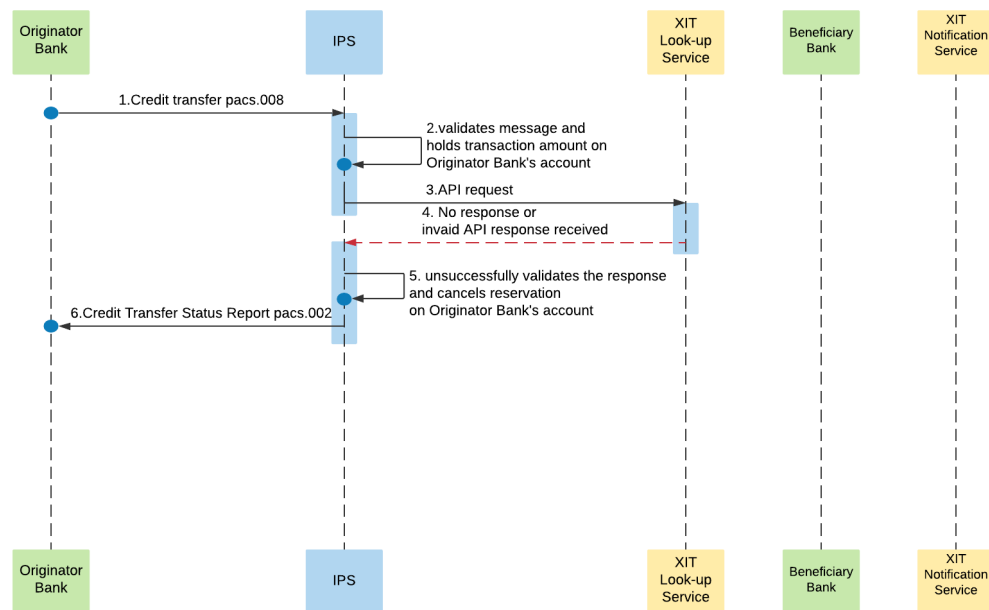
2.1 Credit Transfer validations by IPS are failed (message is invalid), the Originator Bank's net position is not enough to perform Credit Transfer



1. The Originator Bank submits Credit Transfer message pacs.008 to the IPS. This message contains the recipients' name, surname, and phone number instead of the account number of the recipient.
2. The IPS validates the message, checks statuses of the Originator Bank and Beneficiary Bank, checks existing limits for the participants, and checks the IPS net position of the Originator Bank.
3. If these validations failed, then the IPS returns negative Credit Transfer Status Report message pacs.002 to the Originator Bank indicating that the submitted message is invalid or net position is not enough. The Originator Bank may inform its customer (Ultimate Debtor) about the unsuccessful transaction.

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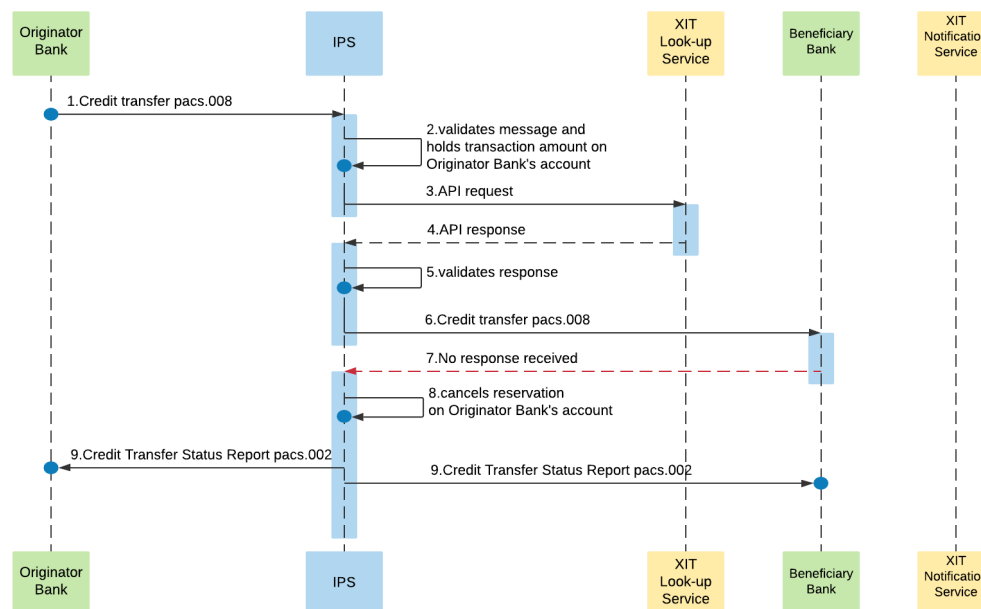
2.2 API call to XIT Look-up Service failed, response from XIT Look-up service is not received or account number/bank name received in response are invalid/incorrect



1. The Originator Bank submits Credit Transfer message pacs.008 to the IPS. This message contains the recipients' name, surname, and phone number instead of the account number of the recipient.
2. The IPS validates the message, checks statuses of the Originator Bank and Beneficiary Bank, checks existing limits for the participants, and checks the IPS net position of the Originator Bank.
3. The IPS initiates an API call to XIT Look-up Service to request the account number associated with the provided phone number and the bank where the account is opened.
4. Unsuccessful scenarios: the XIT Look-up Service is not reachable, or the XIT Look-up Service does not respond, or response received from the XIT Look-up Service or/and response data are invalid.
5. The IPS unsuccessfully validates the response from XIT Look-up Service and cancels reservation on the IPS account of the Originator Bank.
6. The IPS returns negative Credit Transfer Status Report message pacs.002 to the Originator Bank indicating that request to the XIT Look-up Service failed. The Originator Bank may inform its customer (Ultimate Debtor) about the unsuccessful transaction.

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2.3 No response received from the Beneficiary Bank – reject due to timeout



1. The Originator Bank submits Credit Transfer message pacs.008 to the IPS. This message contains the recipients' name, surname, and phone number instead of the account number of the recipient.
2. The IPS validates the message, checks statuses of the Originator Bank and Beneficiary Bank, checks existing limits for the participants, and checks the IPS net position of the Originator Bank. The IPS makes a reservation for the Credit Transfer amount on the IPS account of the Originator Bank.
3. The IPS initiates an API call to XIT Look-up Service to request the account number associated with the provided phone number and the bank where the account is opened.
4. The XIT Look-up Service returns the account number associated with the phone number provided in the request message and the bank where the account is opened.
5. The IPS validated the received response message (whether received account number corresponds to the format standards).
6. The IPS replaces phone number with the corresponding account number received from the XIT Look-up Service and forwards the Credit Transfer message pacs.008 to the Beneficiary Bank.
7. No response (valid) received from the Beneficiary Bank (within the pre-defined time limit).
8. The IPS cancels the reservation made on net position of the Originator Bank's IPS account.
9. The IPS submits negative Credit Transfer Status Report message pacs.002 to the Originator Bank and to the Beneficiary Bank indicating that message is rejected due to timeout. The Originator Bank and the Beneficiary Bank may inform their customers (Ultimate Debtor and Ultimate Creditor accordingly) about the unsuccessful transaction.

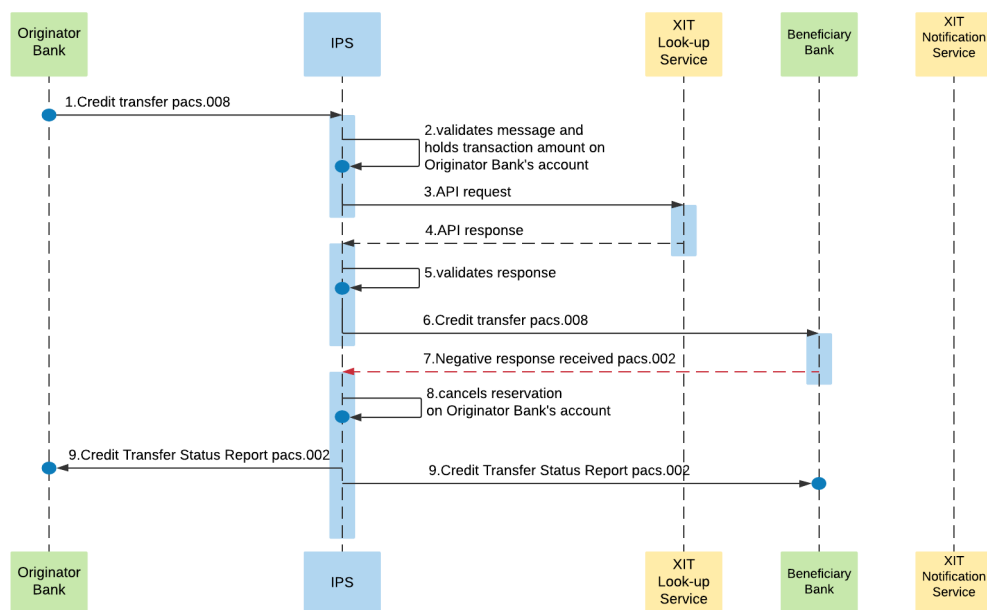
Note: Message exchange between the parties is controlled by the application layer under HTTPs.

If the IPS receives the response from the network that the message Credit Transfer Status Report (pacs.002) is not delivered to the Beneficiary Bank, the IPS is allowed additionally 4

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(four) times to re-submit the Credit Transfer Status Report message to the Beneficiary Bank. If after these 4 attempts the Credit Transfer Status Report (pacs.002) still is not delivered to Beneficiary Bank, before finalizing the credit transaction with the Ultimate Creditor the Beneficiary Bank must initiate Investigation message (pacs.028).

2.4 Credit Transfer validations by the Beneficiary Bank are failed, or it is not possible to credit customer (Ultimate Creditor) due to any reason (account closed, limits are reached etc.)



1. The Originator Bank submits Credit Transfer message pacs.008 to the IPS. This message contains the recipients' name, surname, and phone number instead of the account number of the recipient.
2. The IPS validates the message, checks statuses of the Originator Bank and Beneficiary Bank, checks existing limits for the participants, and checks the IPS net position of the Originator Bank. The IPS makes a reservation for the Credit Transfer amount on the IPS account of the Originator Bank.
3. The IPS initiates an API call to XIT Look-up Service to request the account number associated with the provided phone number and the bank where the account is opened.
4. The XIT Look-up Service returns the account number associated with the phone number provided in the request message and the bank where the account is opened.
5. The IPS validated the received response message (whether received account number corresponds to the format standards).
6. The IPS replaces phone number with the corresponding account number received from the XIT Look-up Service and forwards the Credit Transfer message pacs.008 to the Beneficiary Bank.
7. The Beneficiary Bank receives and validates negatively the Credit Transfer message pacs.008. The Beneficiary Bank responds with the negative Credit Transfer Status Report message pacs.002 indicating the corresponding reason for decline.
8. The IPS cancels the reservation made on net position of the Originator Bank's IPS account.

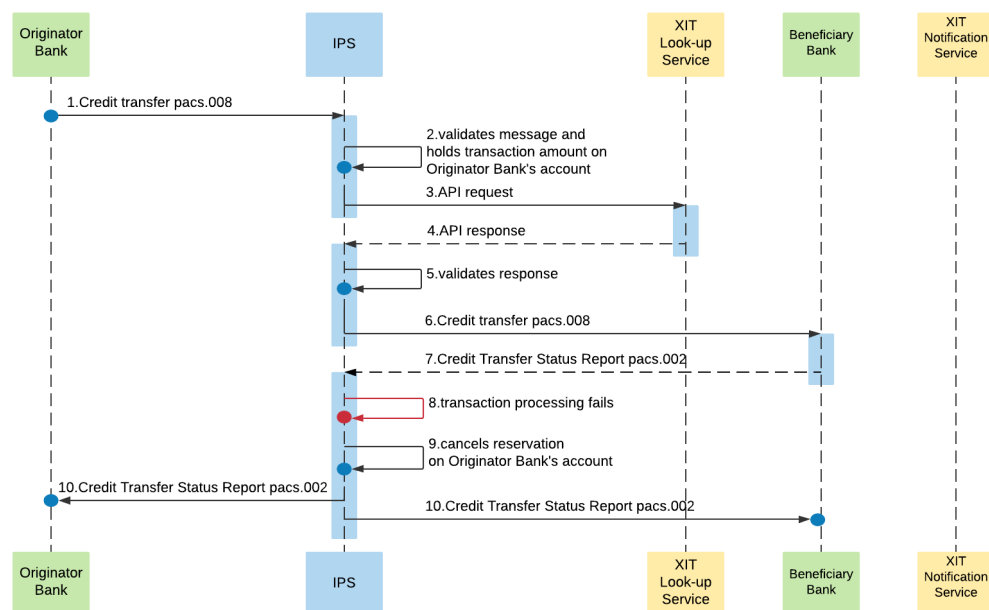
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9. The IPS submits negative Credit Transfer Status Report message pacs.002 to the Originator Bank and to the Beneficiary Bank indicating that message is rejected by the Beneficiary Bank. The Originator Bank and the Beneficiary Bank may inform their customers (Ultimate Debtor and Ultimate Creditor accordingly) about the unsuccessful transaction.

Note: Message exchange between the parties is controlled by the application layer under HTTPs.

If the IPS receives the response from the network that the message Credit Transfer Status Report (pacs.002) is not delivered to the Beneficiary Bank, the IPS is allowed additionally 4 (four) times to re-submit the Credit Transfer Status Report message to the Beneficiary Bank. If after these 4 attempts the Credit Transfer Status Report (pacs.002) still is not delivered to Beneficiary Bank, before finalizing the credit transaction with the Ultimate Creditor the Beneficiary Bank must initiate Investigation message (pacs.028).

2.5 IPS failed to make changes of net position on the participant's accounts



1. The Originator Bank submits Credit Transfer message pacs.008 to the IPS. This message contains the recipients' name, surname, and phone number instead of the account number of the recipient.
2. The IPS validates the message, checks statuses of the Originator Bank and Beneficiary Bank, checks existing limits for the participants, and checks the IPS net position of the Originator Bank. The IPS makes a reservation for the Credit Transfer amount on the IPS account of the Originator Bank.
3. The IPS initiates an API call to XIT Look-up Service to request the account number associated with the provided phone number and the bank where the account is opened.
4. The XIT Look-up Service returns the account number associated with the phone number provided in the request message and the bank where the account is opened.

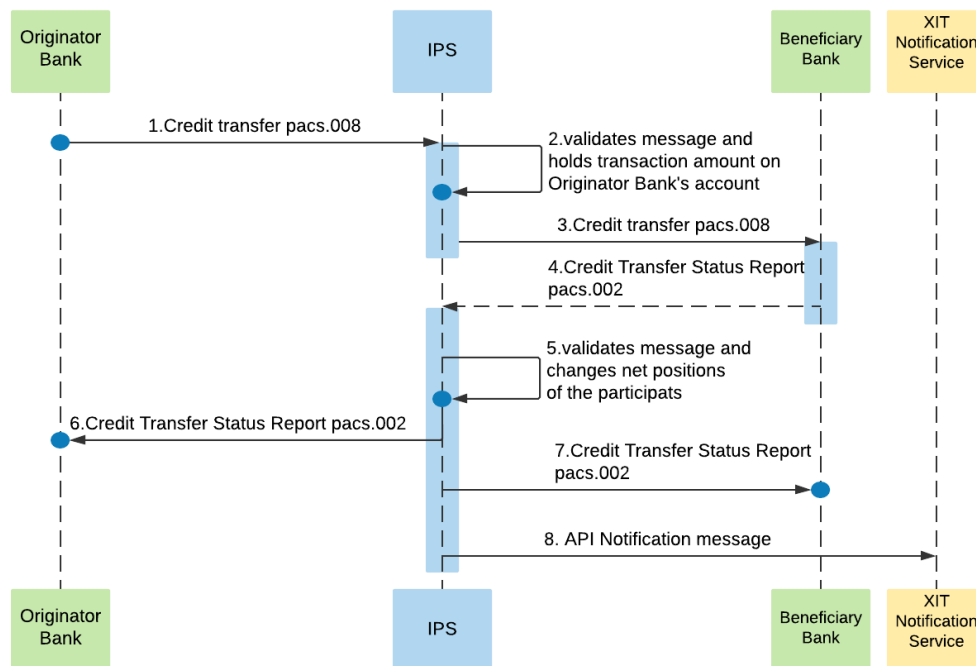
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5. The IPS validated the received response message (whether received account number corresponds to the format standards).
6. The IPS replaces phone number with the corresponding account number received from the XIT Look-up Service and forwards the Credit Transfer message pacs.008 to the Beneficiary Bank.
7. The Beneficiary Bank receives, validates positively the Credit Transfer message pacs.008 and responses with the positive confirmation within the Credit Transfer Status Report message pacs.002.
8. The transaction processing inside the IPS failed due to some reason (mostly technical).
9. The IPS cancels the reservation made on net position of the Originator Bank's IPS account.
10. The IPS submits negative Credit Transfer Status Report message pacs.002 to the Originator Bank and to the Beneficiary Bank indicating that message is rejected due to technical reason (or indicating more specific reason). The Originator Bank and the Beneficiary Bank may inform their customers (Ultimate Debtor and Ultimate Creditor accordingly) about the unsuccessful transaction.

Note: Message exchange between the parties is controlled by the application layer under HTTPs.

If the IPS receives the response from the network that the message Credit Transfer Status Report (pacs.002) is not delivered to the Beneficiary Bank, the IPS is allowed additionally 4 (four) times to re-submit the Credit Transfer Status Report message to the Beneficiary Bank. If after these 4 attempts the Credit Transfer Status Report (pacs.002) still is not delivered to Beneficiary Bank, before finalizing the credit transaction with the Ultimate Creditor the Beneficiary Bank must initiate Investigation message (pacs.028).

3 Credit Transfer to account number without account verification, positive scenario



1. The Originator Bank submits Credit Transfer message pacs.008 to the IPS. This message contains the recipients' name, surname, the account number and bank name of the recipient.
2. The IPS positively validates the message, checks statuses of the Originator Bank and Beneficiary Bank, checks existing limits for the participants, and checks the IPS net position of the Originator Bank. The IPS makes a reservation for the Credit Transfer amount on the IPS account of the Originator Bank.
3. The IPS forwards the Credit Transfer message pacs.008 to the Beneficiary Bank.
4. The Beneficiary Bank receives, validates positively the Credit Transfer message pacs.008 and responds with the positive confirmation within the Credit Transfer Status Report message pacs.002.
5. The IPS positively validates the received response Credit Transfer Status Report message and changes net positions on the participants by moving Credit Transfer amount (including fee, if applicable) from the Originator Bank's IPS account to Beneficiary Bank's IPS account.
6. The IPS submits Credit Transfer Status Report message pacs.002 to the Originator Bank. Based on this message the Originator Bank may inform its customer (Ultimate Debtor) about the successful transaction.
7. The IPS submits Credit Transfer Status Report message pacs.002 to the Beneficiary Bank. Based on this message the Beneficiary Bank may finalize the transaction with its customer (Ultimate Creditor) by making the Credit Transfer amount available on the customers' account.

Note: Message exchange between the parties is controlled by the application layer under HTTPs.

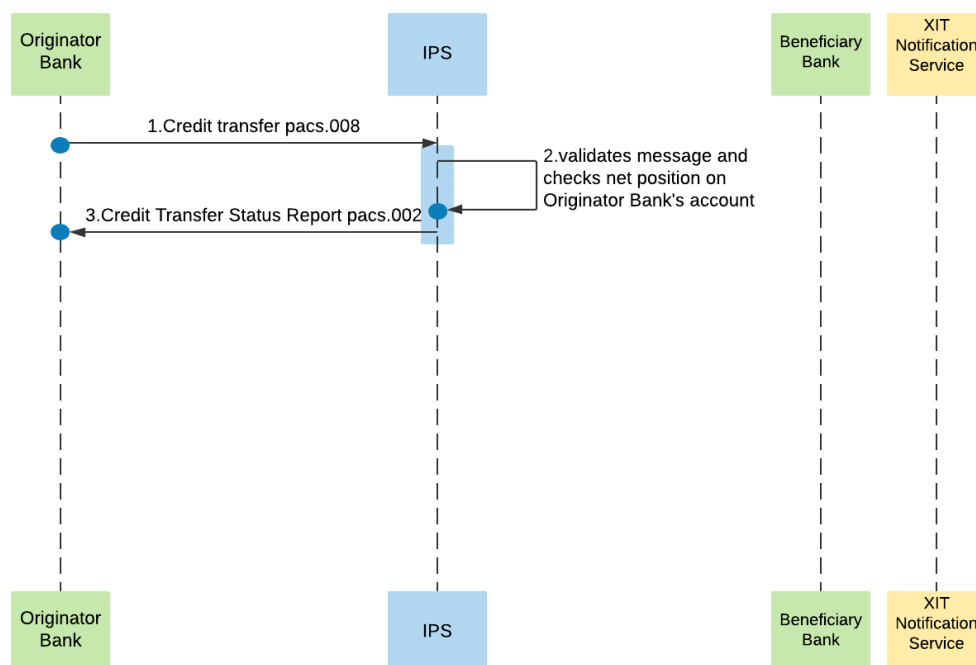
If the IPS receives the response from the network that the message Credit Transfer Status Report (pacs.002) is not delivered to the Beneficiary Bank, the IPS is allowed additionally 4 (four) times

to re-submit the Credit Transfer Status Report message to the Beneficiary Bank. If after these 4 attempts the Credit Transfer Status Report (pacs.002) still is not delivered to Beneficiary Bank, before finalizing the credit transaction with the Ultimate Creditor the Beneficiary Bank must initiate Investigation message (pacs.028).

8. Additionally, after successful transaction processing the IPS generate a notification API message for XIT Notification Service for further SMS delivery to the customer (Ultimate Creditor).

4 Credit Transfer to account number without account verification, negative scenarios

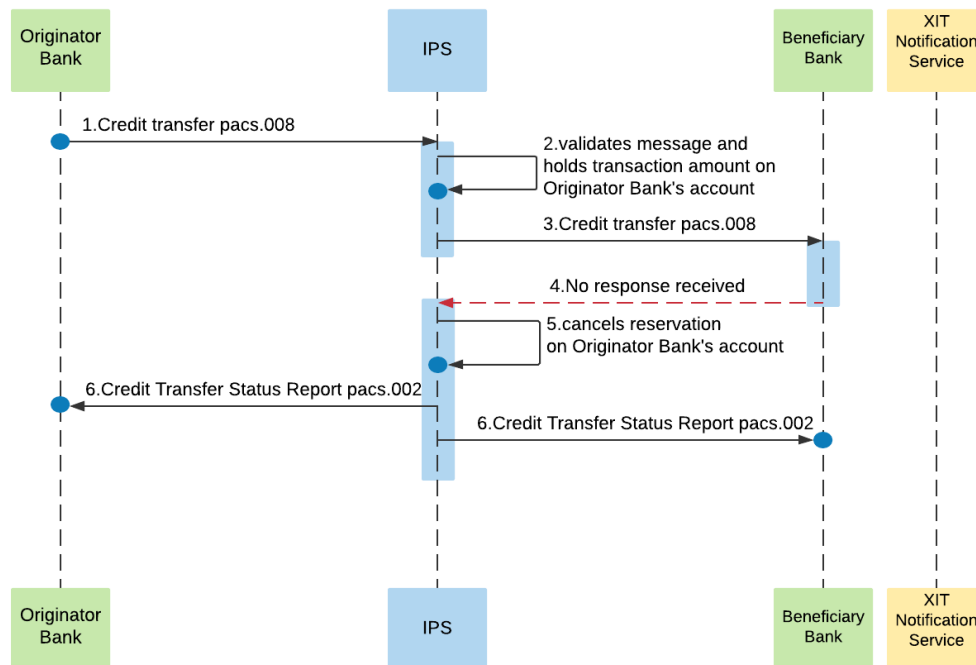
4.1 Credit Transfer validations by IPS are failed (message is invalid), the Originator Bank's net position is not enough to perform Credit Transfer



1. The Originator Bank submits Credit Transfer message pacs.008 to the IPS. This message contains the recipients' name, surname, the account number and bank name of the recipient.
2. The IPS validates the message, checks statuses of the Originator Bank and Beneficiary Bank, checks existing limits for the participants, and checks the IPS net position of the Originator Bank.
3. If these validations failed, then the IPS returns negative Credit Transfer Status Report message pacs.002 to the Originator Bank indicating that the submitted message is invalid or net position is not enough. The Originator Bank may inform its customer (Ultimate Debtor) about the unsuccessful transaction.

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4.2 No response received from the Beneficiary Bank – reject due to timeout

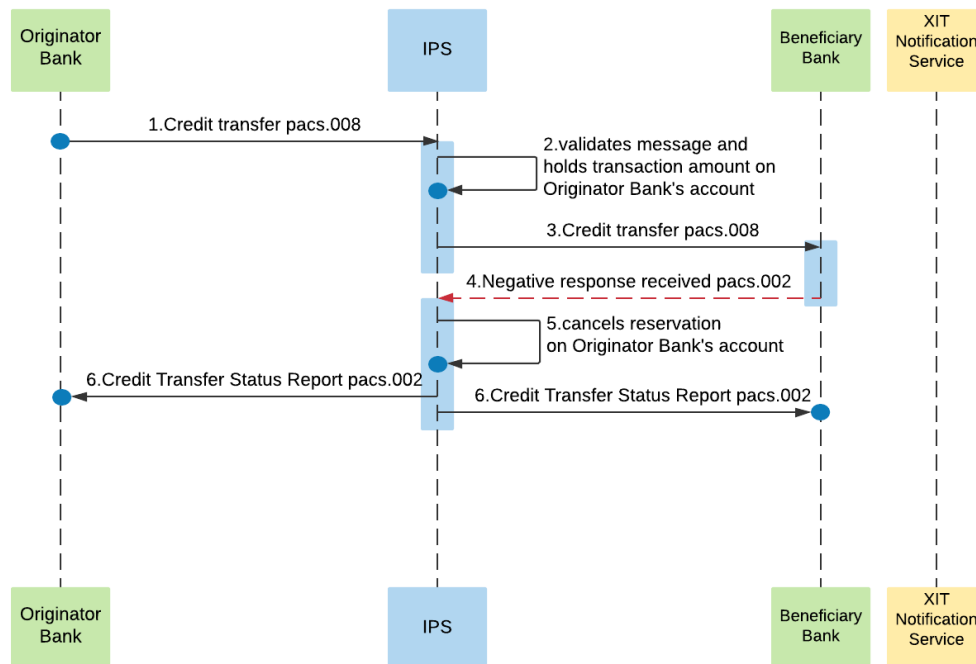


1. The Originator Bank submits Credit Transfer message pacs.008 to the IPS. This message contains the recipients' name, surname, the account number and bank name of the recipient.
2. The IPS validates the message, checks statuses of the Originator Bank and Beneficiary Bank, checks existing limits for the participants, and checks whether the Originator Bank's net position allows performing the transaction. The IPS makes a reservation for the Credit Transfer amount on the IPS account of the Originator Bank.
3. The IPS forwards the Credit Transfer message pacs.008 to the Beneficiary Bank.
4. No response (valid) received from the Beneficiary Bank (within the pre-defined time limit).
5. The IPS cancels the reservation made on net position of the Originator Bank's IPS account.
6. The IPS submits negative Credit Transfer Status Report message pacs.002 to the Originator Bank and to the Beneficiary Bank indicating that message is rejected due to timeout.
The Originator Bank and the Beneficiary Bank may inform their customers (Ultimate Debtor and Ultimate Creditor accordingly) about the unsuccessful transaction.

Note: Message exchange between the parties is controlled by the application layer under HTTPs.

If the IPS receives the response from the network that the message Credit Transfer Status Report (pacs.002) is not delivered to the Beneficiary Bank, the IPS is allowed additionally 4 (four) times to re-submit the Credit Transfer Status Report message to the Beneficiary Bank. If after these 4 attempts the Credit Transfer Status Report (pacs.002) still is not delivered to Beneficiary Bank, before finalizing the credit transaction with the Ultimate Creditor the Beneficiary Bank must initiate Investigation message (pacs.028).

4.3 Credit Transfer validations by the Beneficiary Bank are failed, it is not possible to credit customer (Ultimate Creditor) due to any reason (account closed, limits are reached etc.)



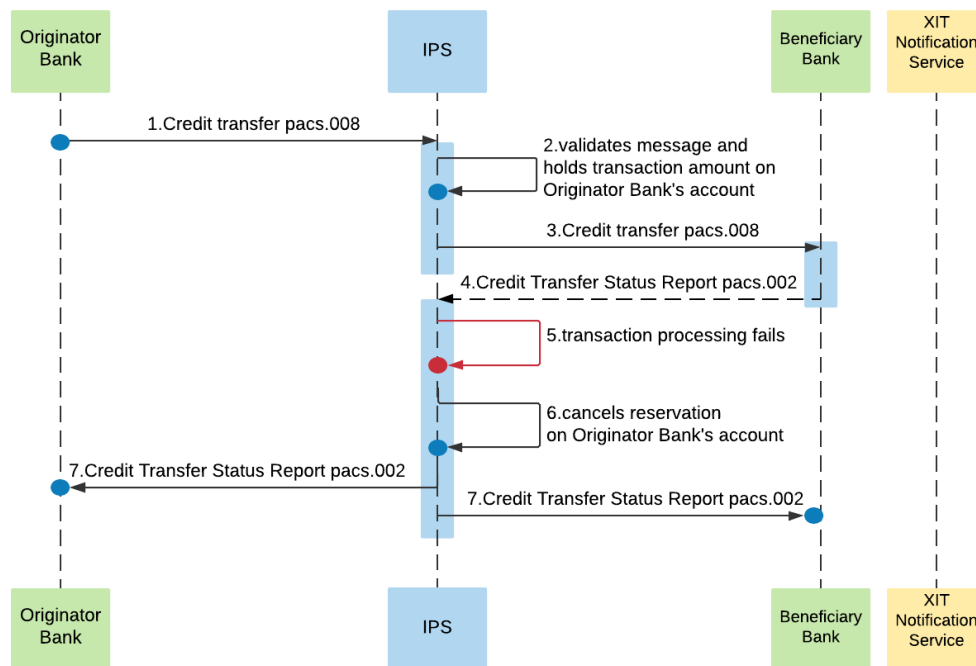
1. The Originator Bank submits Credit Transfer message pacs.008 to the IPS. This message contains the recipients' name, surname, the account number and bank name of the recipient.
2. The IPS validates the message, checks statuses of the Originator Bank and Beneficiary Bank, checks existing limits for the participants, and checks whether the Originator Bank's net position allows performing the transaction. The IPS makes a reservation for the Credit Transfer amount on the IPS account of the Originator Bank.
3. The IPS forwards the Credit Transfer message pacs.008 to the Beneficiary Bank.
4. The Beneficiary Bank receives and validates negatively the Credit Transfer message pacs.008. The Beneficiary Bank responds with the negative Credit Transfer Status Report message pacs.002 indicating the corresponding reason for decline.
5. The IPS cancels the reservation made on net position of the Originator Bank's IPS account.
6. The IPS submits negative Credit Transfer Status Report message pacs.002 to the Originator Bank and to the Beneficiary Bank indicating that message is rejected by the Beneficiary Bank. The Originator Bank and the Beneficiary Bank may inform their customers (Ultimate Debtor and Ultimate Creditor accordingly) about the unsuccessful transaction.

Note: Message exchange between the parties is controlled by the application layer under HTTPs.

If the IPS receives the response from the network that the message Credit Transfer Status Report (pacs.002) is not delivered to the Beneficiary Bank, the IPS is allowed additionally 4 (four) times to re-submit the Credit Transfer Status Report message to the Beneficiary Bank. If after these 4 attempts the Credit Transfer Status Report (pacs.002) still is not delivered to Beneficiary Bank,

before finalizing the credit transaction with the Ultimate Creditor the Beneficiary Bank must initiate Investigation message (pacs.028).

4.4 IPS failed to make changes of net position on the participant's accounts



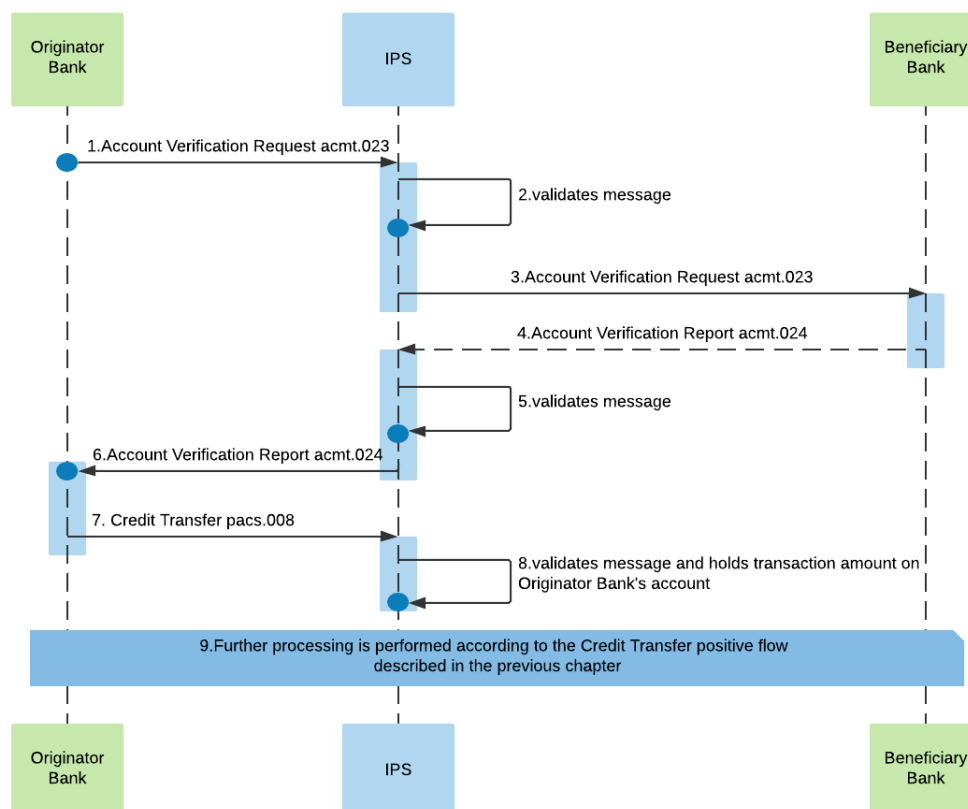
1. The Originator Bank submits Credit Transfer message pacs.008 to the IPS. This message contains the recipients' name, surname, the account number and bank name of the recipient.
2. The IPS validates the message, checks statuses of the Originator Bank and Beneficiary Bank, checks existing limits for the participants, and checks whether the Originator Bank's net position allows performing the transaction. The IPS makes a reservation for the Credit Transfer amount on the IPS account of the Originator Bank.
3. The IPS forwards the Credit Transfer message pacs.008 to the Beneficiary Bank.
4. The Beneficiary Bank receives, validates positively the Credit Transfer message pacs.008 and responds with the positive confirmation within the Credit Transfer Status Report message pacs.002.
5. The transaction processing inside the IPS failed due to some reason (mostly technical).
6. The IPS cancels the reservation made on net position of the Originator Bank's IPS account.
7. The IPS submits negative Credit Transfer Status Report message pacs.002 to the Originator Bank and to the Beneficiary Bank indicating that message is rejected due to technical reason (or indicating more specific reason). The Originator Bank and the Beneficiary Bank may inform their customers (Ultimate Debtor and Ultimate Creditor accordingly) about the unsuccessful transaction.

Note: Message exchange between the parties is controlled by the application layer under HTTPs.

If the IPS receives the response from the network that the message Credit Transfer Status Report (pacs.002) is not delivered to the Beneficiary Bank, the IPS is allowed additionally 4 (four) times to

re-submit the Credit Transfer Status Report message to the Beneficiary Bank. If after these 4 attempts the Credit Transfer Status Report (pacs.002) still is not delivered to Beneficiary Bank, before finalizing the credit transaction with the Ultimate Creditor the Beneficiary Bank must initiate Investigation message (pacs.028).

5 Credit Transfer to account with account verification, positive scenario

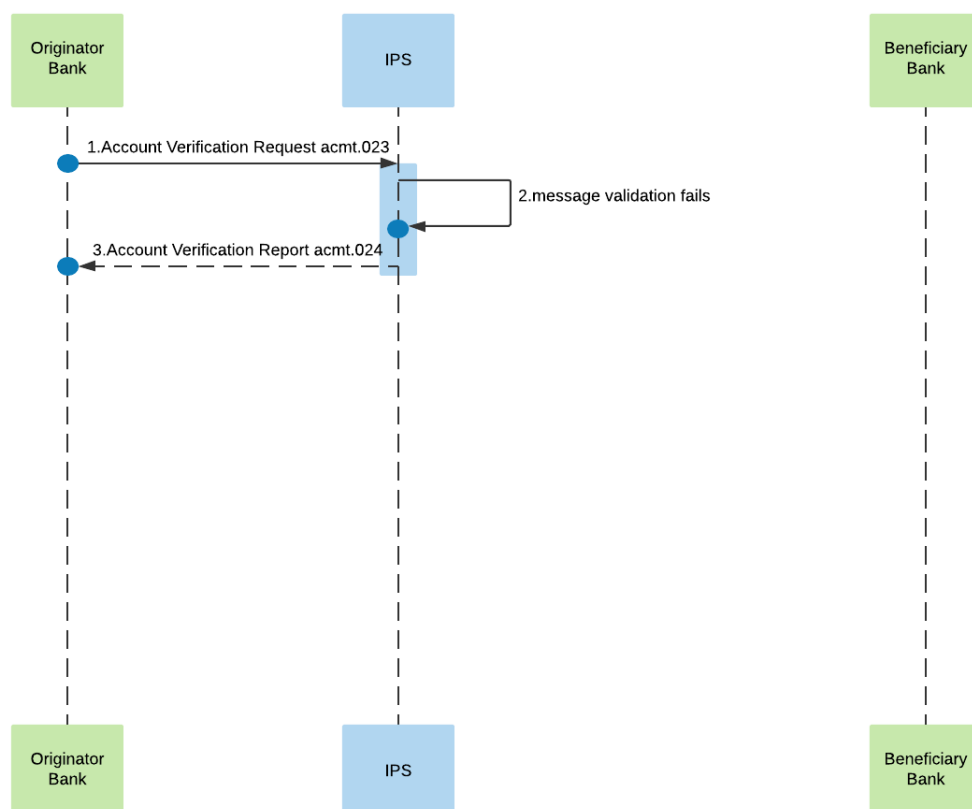


1. When Originator Bank's customer (Ultimate Debtor) initiates Credit Transfer to the account, first the Originator Bank must submit Account Verification Request acmt.023 to the IPS. The request contains recipient's account number and the Beneficiary Bank details.
2. The IPS positively validates the Account Verification Request acmt.023.
3. The IPS forwards Account Verification Request message acmt.023 to the Beneficiary Bank.
4. The Beneficiary Bank returns Account Verification Report message acmt.024, containing the recipient's (Ultimate Creditor) name and surname, account number (Account number is not required (optional) to be returned since that was what was used initially).
5. The IPS positively validates received Report message.
6. The IPS forwards Response (result) of Account Verification acmt.024 to the Originator Bank. The Originator Bank has to communicate received data (name and surname of Ultimate Creditor) with its customer (Ultimate Debtor) and to get confirmation, that recipients' data is correct. If confirmation from the payment initiator is received, the Originator Bank is allowed to proceed with the Credit Transfer request submitting to the IPS.

7. The Originator Bank submits the Credit Transfer message pacs.008 to The IPS. This message contains the recipients' name, surname, bank name and the account number of the recipient.
8. The IPS validates the message, checks statuses of the Originator Bank and Beneficiary Bank, checks existing limits for the participants, and checks whether the Originator Bank's net position allows performing the transaction. The IPS makes a reservation for the Credit Transfer amount on the IPS account of the Originator Bank.
9. Further processing is performed according to the Credit Transfer to Account number without Account Verification positive flow described in the previous chapter.

6 Credit Transfer to account with Account Verification negative scenarios

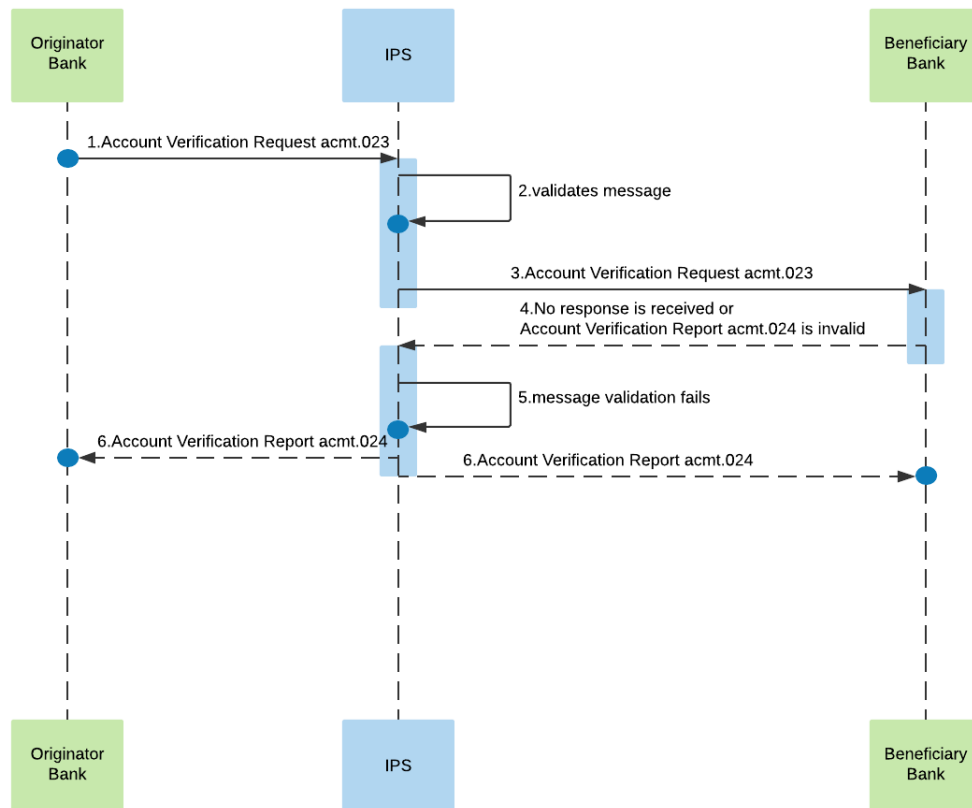
6.1 Account Verification Request message validations by IPS are failed (message is invalid)



1. The Originator Bank submits Account Verification Request acmt.023 to the IPS. The request contains recipient's account number.
2. The IPS negatively validates the Account Verification Request acmt.023.
3. The IPS generates Account Verification Report message acmt.024 to the Originator Bank indicating that submitted Account Verification Request message was invalid.

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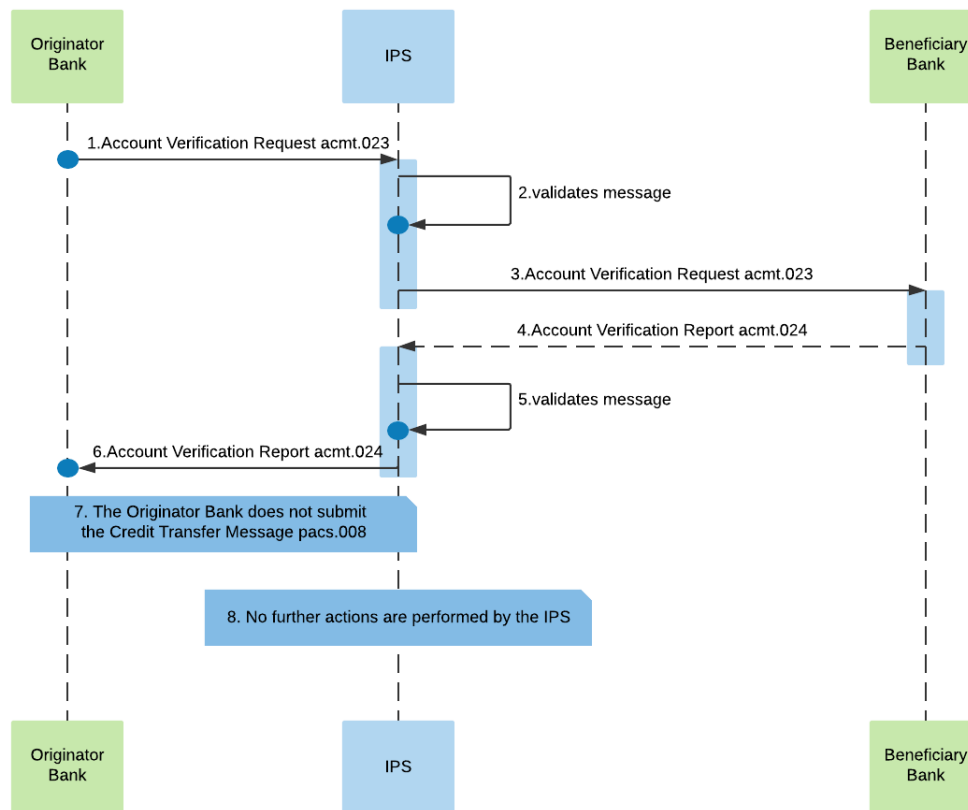
6.2 The Beneficiary Bank is not responding, or Account Verification Report message is invalid



1. The Originator Bank submits Account Verification Request acmt.023 to the IPS. The request contains recipient's account number.
2. The IPS positively validates the Account Verification Request acmt.023.
3. The IPS forwards Account Verification Request message acmt.023 to the Beneficiary Bank.
4. The Beneficiary Bank does not respond or returns invalid Account Verification Report message acmt.024.
5. The IPS negatively validates received Report message.
6. The IPS generates Account Verification Report message acmt.024 to the Originator Bank and to the Beneficiary Bank indicating that submitted Account Verification Request message was invalid

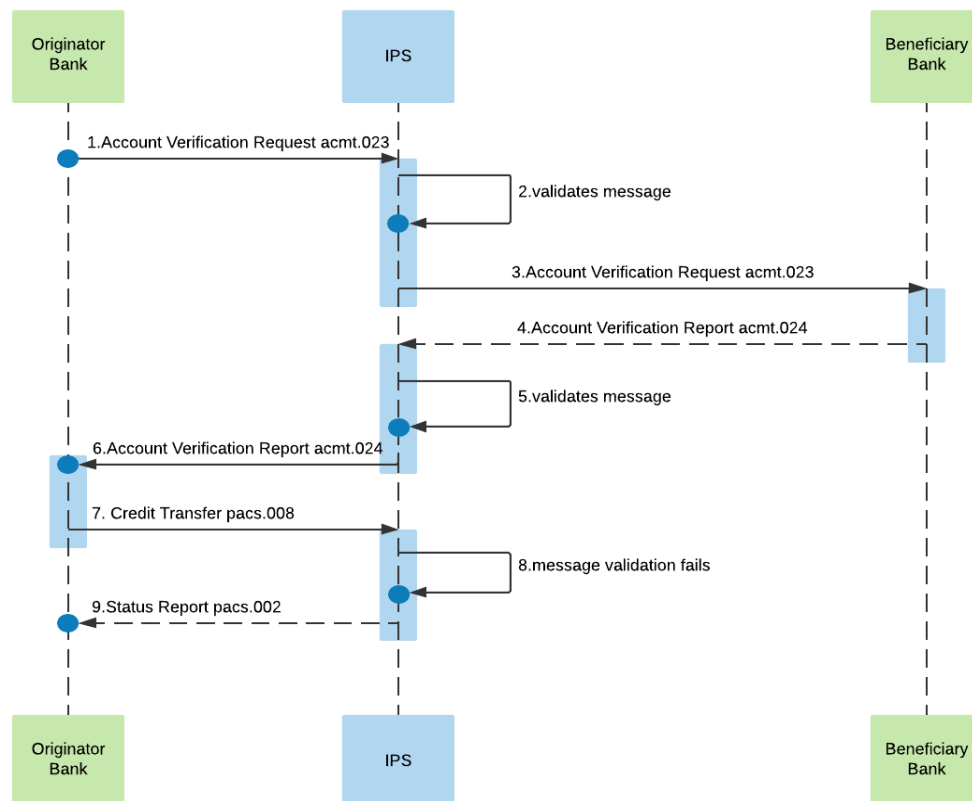
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6.3 The Originator Bank does not submit the Credit Transfer message



1. The Originator Bank submits Account Verification Request acmt.023 to the IPS. The request contains recipient's account number.
2. The IPS positively validates the Account Verification Request acmt.023.
3. The IPS forwards Account Verification Request message acmt.023 to the Beneficiary Bank.
4. The Beneficiary Bank returns Account Verification Report message acmt.024, containing the recipient's (Ultimate Creditor) name and surname.
5. The IPS positively validates received Report message.
6. The IPS forwards Response (result) of Account Verification acmt.024 to the Originator Bank. The Originator Bank has to communicate received data (name and surname of Ultimate Creditor) with its customer (Ultimate Debtor) and to get confirmation, that recipients' data is correct. If confirmation from the payment initiator is received, the Originator Bank is allowed to proceed with the Credit Transfer request submitting to the IPS.
7. The Originator Bank does not submit the Credit Transfer message pacs.008 to Tieto IPS.
8. No further actions are performed by the IPS.

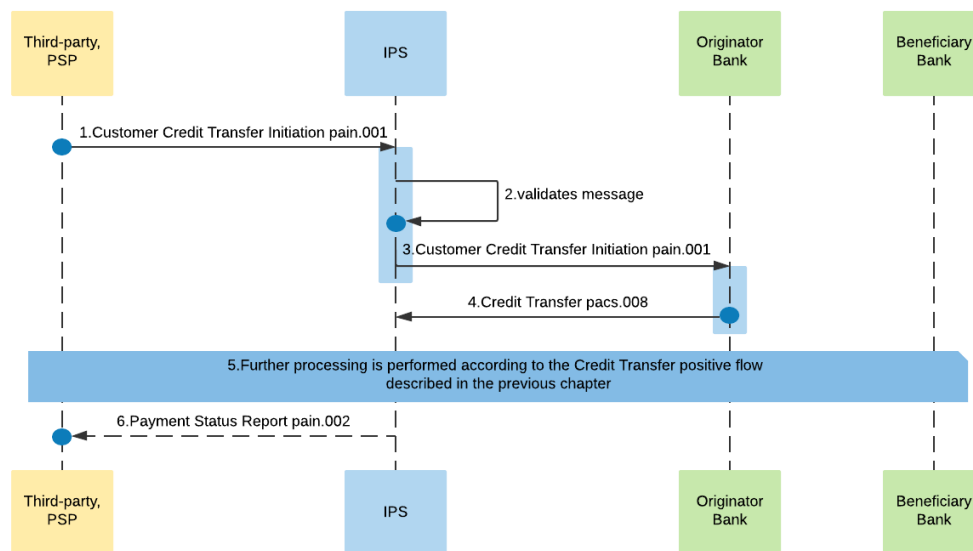
6.4 Credit Transfer validations by IPS are failed (message is invalid), or the Originator Bank's net position is not enough to perform Credit Transfer



1. The Originator Bank submits Account Verification Request acmt.023 to the IPS. The request contains recipient's account number.
2. The IPS positively validates the Account Verification Request acmt.023.
3. The IPS forwards Account Verification Request message acmt.023 to the Beneficiary Bank.
4. The Beneficiary Bank returns Account Verification Report message acmt.024, containing the recipient's (Ultimate Creditor) name and surname.
5. The IPS positively validates received Report message.
6. The IPS forwards Response (result) of Account Verification acmt.024 to the Originator Bank. The Originator Bank has to communicate received data (name and surname of Ultimate Creditor) with its customer (Ultimate Debtor) and to get confirmation, that recipients' data is correct. If confirmation from the payment initiator is received, the Originator Bank is allowed to proceed with the Credit Transfer request submitting to the IPS.
7. The Originator Bank submits the Credit Transfer message pacs.008 to The IPS. This message contains the recipients' name, surname, bank name and the account number of the recipient.
8. The IPS validates the message, checks statuses of the Originator Bank and Beneficiary Bank, checks existing limits for the participants, and checks whether the Originator Bank's net position allows performing the transaction.
9. If these validations failed, then the IPS returns negative Credit Transfer Status Report message pacs.002 to the Originator Bank indicating that the submitted message is invalid or net position

is not enough. The Originator Bank may inform its customer (Ultimate Debtor) about the unsuccessful transaction.

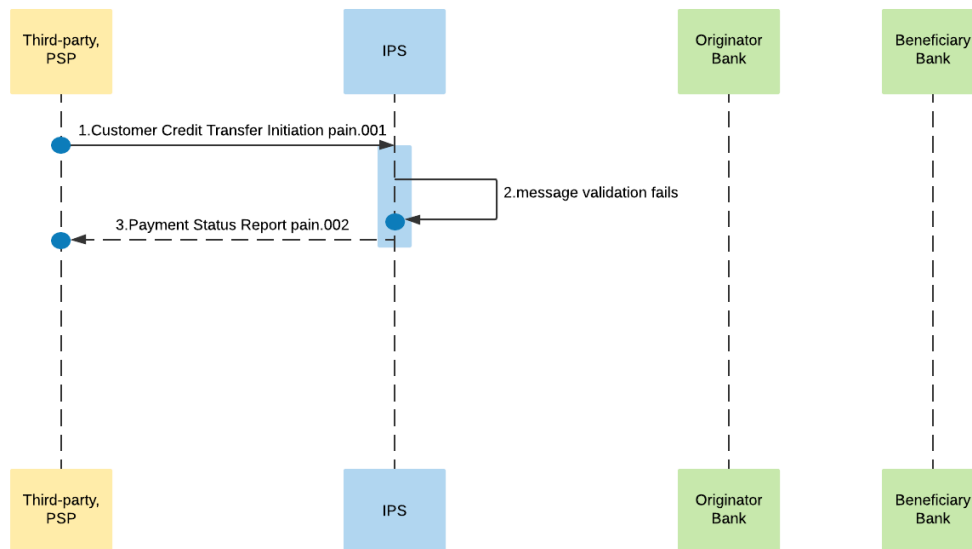
7 Request to Pay by PSP, positive scenario



1. The IPS solution supports the possibility to initiate a Credit Transfer request by Third-party PSP. The Third-party PSP (for example, via IPSL PGW) submits Customer Credit Transfer initiation message pain.001 to the IPS. The message contains data necessary for payment initiation: Originator Bank name, Beneficiary Bank name, Ultimate Debtor's name and account number, Ultimate Creditor's name and account number, payment details.
2. The IPS system validates the payment initiation message.
3. The IPS forwards Customer Credit Transfer initiation message pain.001 to the Originator Bank. The Originator Bank has to communicate the request with its customer (Ultimate Debtor) and to receive a confirmation for the Credit Transfer initiation. The communication between the Originator Bank and the Ultimate Debtor is out of the scope of this implementation and needs to be developed by the Originator Bank outside the IPS solution.
4. If the Originator Bank's customer (Ultimate Debtor) confirms the payment, the Originator Bank initiates the Credit Transfer message pacs.008.
5. Further processing is performed according to the Credit Transfer to Account number positive flow described in the previous chapter. If necessary, the Account Verification procedure might be performed (IPSL to decide in which cases Account Verification procedure is needed).
6. After the Credit Transfer is successfully processed, the IPS initiates Payment Status Report message pain.002 to inform the Third-party PSP about the final status of the initiated payment.

8 Request to Pay by PSP, negative scenarios

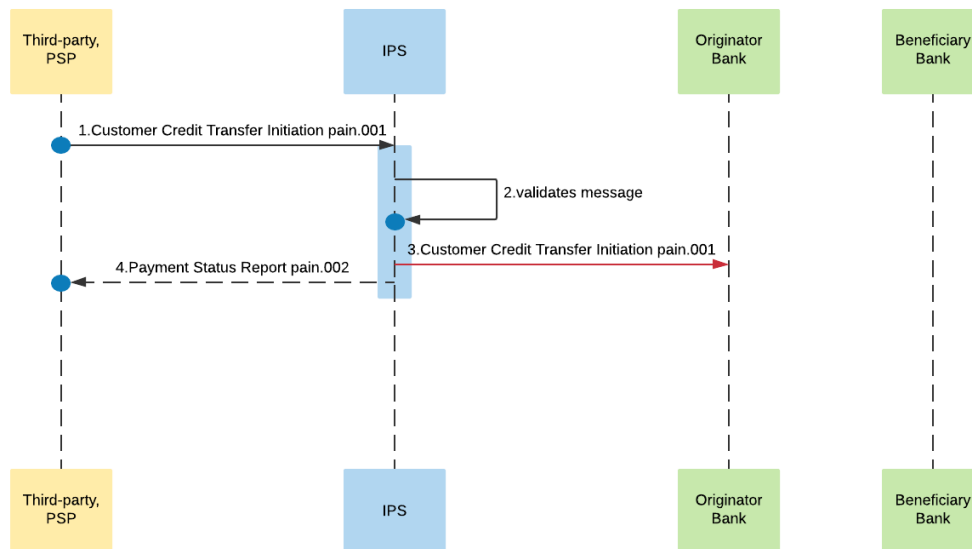
8.1 Customer Credit Transfer Initiation message validations by IPS are failed (message is invalid)



1. The Third-party PSP submits Customer Credit Transfer initiation message pain.001 to the IPS. The message contains data necessary for payment initiation: Originator Bank name, Beneficiary Bank name, Ultimate Debtor's name and account number (or phone number), Ultimate Creditor's name and account number (or phone number), payment details.
2. The IPS system unsuccessfully validates the payment initiation message.
3. The IPS returns rejection message Payment Status Report pain.002 indicating the reason of decline of the Customer Credit Transfer initiation message.

IPSL

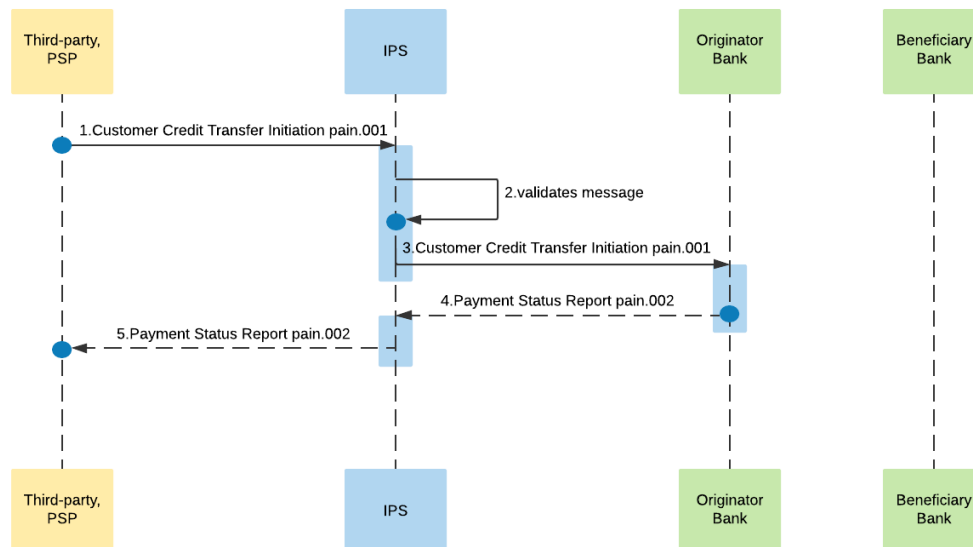
8.2 It is not possible to send Customer Credit Transfer Initiation message to the Originator Bank



1. The Third-party PSP submits Customer Credit Transfer initiation message pain.001 to the IPS. The message contains data necessary for payment initiation: Originator Bank name, Beneficiary Bank name, Ultimate Debtor's name and account number (or phone number), Ultimate Creditor's name and account number (or phone number), payment details.
2. The IPS system successfully validates the payment initiation message.
3. The IPS is trying to forward the Customer Credit Transfer initiation message pain.001 to the Originator Bank. The originator Bank is not reachable.
4. The IPS returns rejection message Payment Status Report pain.002 indicating the reason of decline of the Customer Credit Transfer initiation message.

IPSL

8.3 The Request to Pay is rejected by the Originator Bank (by bank, by customer) or the Originator Bank is not responding



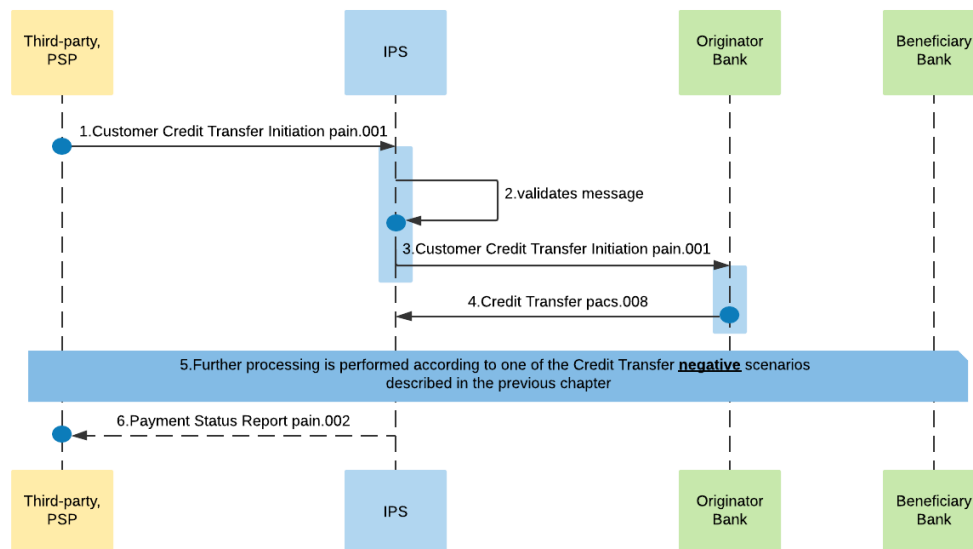
1. The Third-party PSP submits Customer Credit Transfer initiation message pain.001 to the IPS. The message contains data necessary for payment initiation: Originator Bank name, Beneficiary Bank name, Ultimate Debtor's name and account number (or phone number), Ultimate Creditor's name, and account number (or phone number), payment details.
2. The IPS system successfully validates the payment initiation message.
3. The IPS forwards Customer Credit Transfer initiation message pain.001 to the Originator Bank. The Originator Bank has to communicate the request with its customer (Ultimate Debtor) and to receive a confirmation for the Credit Transfer initiation.
4. If the Originator Bank decides to reject received request to pay, it responds with the Payment Status Report pain.002 message.
5. The IPS returns rejection message Payment Status Report pain.002 to the Third-party PSP indicating the reason for the decline of the Customer Credit Transfer initiation message.

In case there is no response received from the Originator Bank (within the predefined time limit, if applicable), the IPS also informs the Third-party PSP about the rejection of the initial request to pay (pain.002).

In case the received Payment Status Report (pain.002) message is invalid, the IPS returned negative response to the Originator Bank within the submits Payment Status Report (pain.002) message. And informs the Third-party PSP about the rejection of the initial request to pay by submitting Payment Status Report pain.002 message.

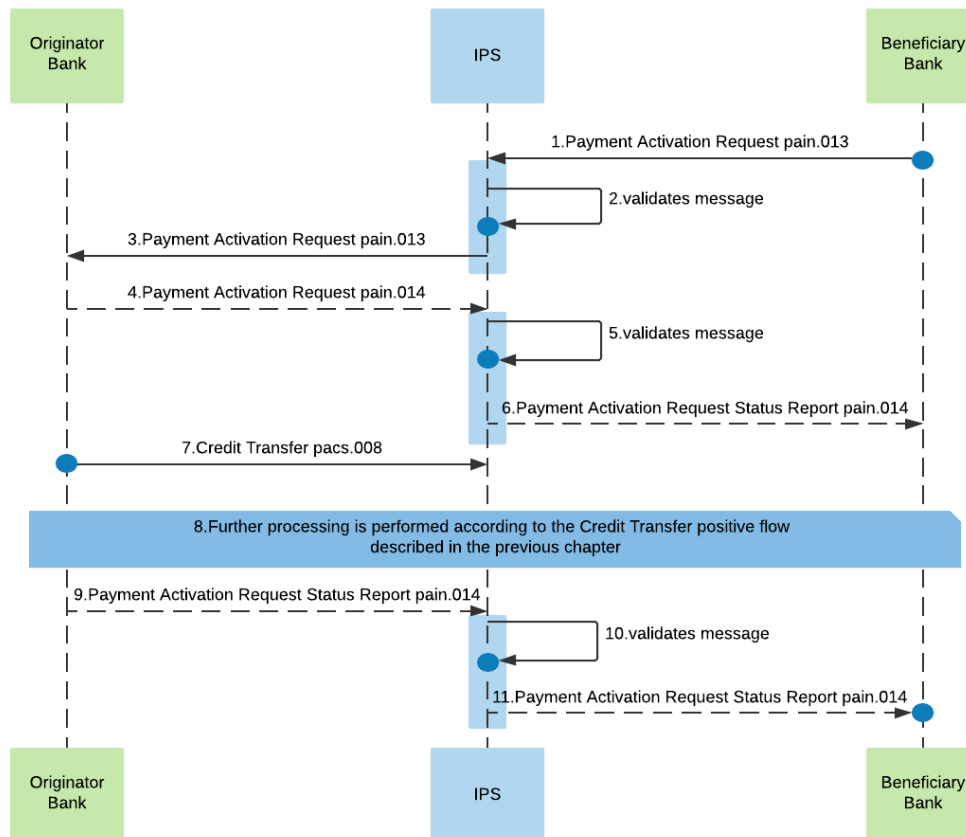
IPSL

8.4 The Credit Transfer processing fails



1. The Third-party PSP submits Customer Credit Transfer initiation message pain.001 to the IPS. The message contains data necessary for payment initiation: Originator Bank name, Beneficiary Bank name, Ultimate Debtor's name and account number (or phone number), Ultimate Creditor's name, and account number (or phone number), payment details.
2. The IPS system successfully validates the payment initiation message.
3. The IPS forwards Customer Credit Transfer initiation message pain.001 to the Originator Bank. The Originator Bank has to communicate the request with its customer (Ultimate Debtor) and to receive a confirmation for the Credit Transfer initiation.
4. If the Originator Bank's customer (Ultimate Debtor) confirms the payment, the Originator Bank initiates the Credit Transfer message pacs.008.
5. Further processing is performed according to one of the Credit Transfer to Account number negative scenarios described in the previous chapters.
6. After the IPS ascertains that Credit Transfer message processing failed, The IPS submits rejection message Payment Status Report pain.002 to the Third-party PSP to inform it about the decline of the Customer Credit Transfer initiation message.

9 Request to Pay, positive scenario



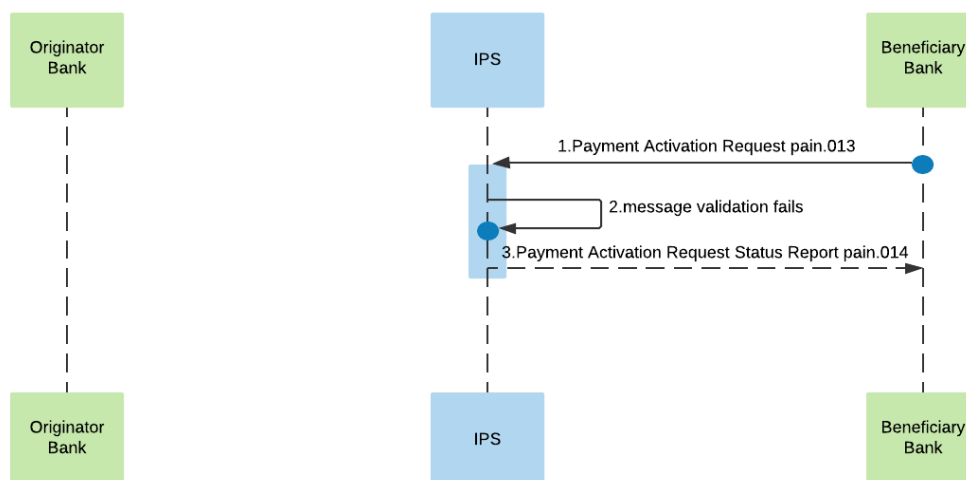
The IPS ensures possibility for participants to request credit payment by using Payment Activation Request message (pain.013), which is sent by the Beneficiary Bank to the Originator Bank through the IPS system. As this is a Request for Payment, it does not automatically follow by a traditional Credit Transfer (pacs.008) message flow where the Originator Bank is the sending party and the Beneficiary Bank is the receiving party. If Request for Payment (pain.013) is approved by the Ultimate Debtor, the Originator Bank generates Payment Activation Request Status Report (pain.014) with positive response and the Originator Bank must initiate IPS Credit Transfer message (pacs.008). The IPS does not wait for the Credit Transfer message initiation and does not control this step within the timeout procedure.

1. The Beneficiary Bank initiates Payment Activation Request pain.013 to request the Originator Bank to execute credit transfer from the Ultimate Debtor in the favour of the Ultimate Creditor. The request message contains the Originator Bank name, Beneficiary Bank name, Ultimate Debtor's name and account number, Ultimate Creditor's name, and account number, payment details.
2. The IPS successfully validates the received message.
3. The IPS forwards Payment Activation Request pain.013 to The Originator Bank, where Ultimate Debtor (addressee of the request) needs to confirm his or her willingness to transfer money to the requestor. Only after positive confirmation is received, the Originator Bank is allowed to proceed with the Credit Transfer request submitting to the IPS.

4. If confirmation from the Ultimate Debtor is received, The Originator Bank submits Payment Activation Request Status (pain.014) with a positive result of the request to pay.
5. The IPS successfully validates the received message.
6. The IPS forwards the Payment Activation Request Status Report (pain.014) to the Beneficiary Bank.
7. The Originator Bank Initiates Credit Transfer message pacs.008 to the IPS.
8. Further processing is performed according to the Credit Transfer to Account number positive flow described in the previous chapter. If necessary, the Account Verification procedure might be performed (IPSL to decide in which cases Account Verification procedure is needed).
9. The Originator Bank may initiate the new Payment Activation Status Report pain.014 to inform the Beneficiary Bank about the final status of the transaction chain caused by the initial Credit Payment Activation Request message submitted by Beneficiary Bank, including the status of Credit Transfer pacs.008, if Credit Payment Activation Request was positively accepted. This step is optional!
10. The IPS successfully validates the received message.
11. The IPS forwards Payment Activation Status Report pain.014 to the Beneficiary Bank.

10 Request to Pay, negative scenarios

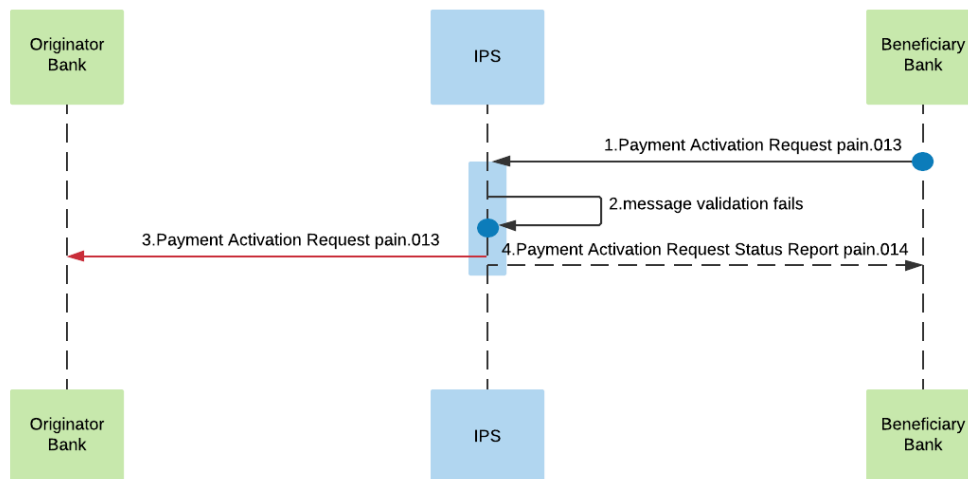
10.1 Customer Credit Transfer Initiation message validations by IPS are failed (message is invalid)



1. The beneficiary Bank submits Payment Activation Request pain.013 to the IPS. The message contains data necessary for payment initiation: Originator Bank name, Beneficiary Bank name, Ultimate Debtor's name and account number (or phone number), Ultimate Creditor's name and account number (or phone number), payment details.
2. The IPS system unsuccessfully validates the payment initiation message.
3. The IPS returns rejection message Payment Activation Request Status Report pain.014 indicating the reason of decline of the Payment Activation Request message.

IPSL

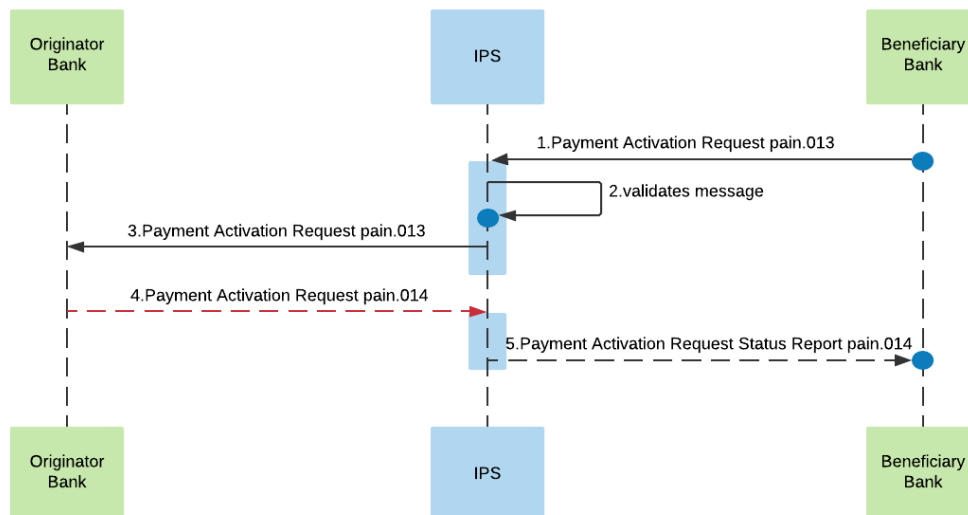
10.2 It is not possible to send Customer Credit Transfer Initiation message to the Originator Bank



1. The Beneficiary Bank submits Payment Activation Request pain.013 to the IPS. The message contains data necessary for payment initiation: Originator Bank name, Beneficiary Bank name, Ultimate Debtor's name and account number (or phone number), Ultimate Creditor's name and account number (or phone number), payment details.
2. The IPS system successfully validates the payment initiation message.
3. The IPS is trying to forward the Payment Activation Request message pain.013 to the Originator Bank. The originator Bank is not reachable.
4. The IPS returns rejection message Payment Activation Request Status Report pain.014 indicating the reason of decline of the Payment Activation Request message.

IPSL

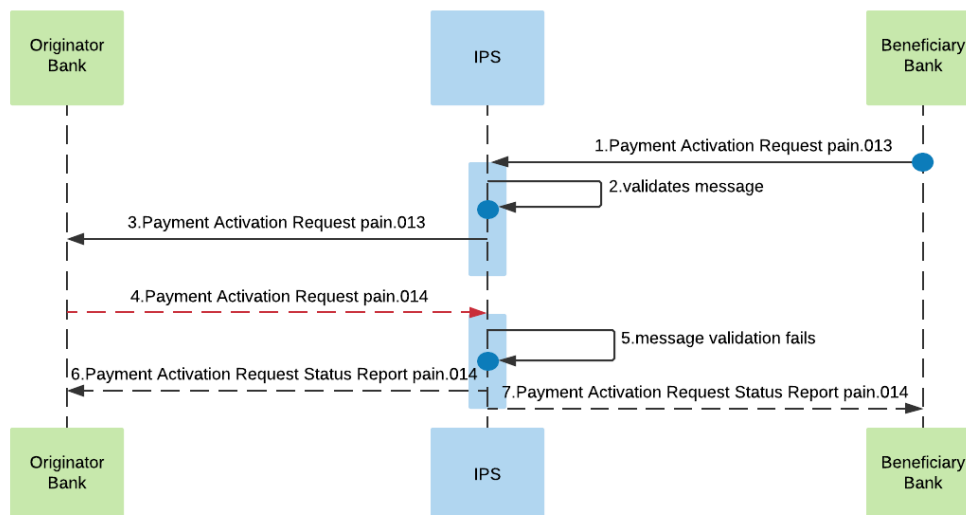
10.3 The Request to Pay is rejected (by bank, by customer) or Originator Bank is not responding.



1. The beneficiary Bank submits Payment Activation Request pain.013 to the IPS. The message contains data necessary for payment initiation: Originator Bank name, Beneficiary Bank name, Ultimate Debtor's name and account number (or phone number), Ultimate Creditor's name and account number (or phone number), payment details.
2. The IPS system successfully validates the payment initiation message.
3. The IPS forwards the Payment Activation Request message pain.013 to the Originator Bank the Ultimate Debtor (addressee of the request) needs to confirm his or her willingness to transfer money to the requestor.
4. If the Originator Bank rejects received request to pay, it responds with the negative Payment Activation Request Status Report pain.014 message indicating the reason of the reject.
5. The IPS forwards rejection message Payment Activation Request Status Report message pain.014 to the Beneficiary Bank indicating the reason for the decline of the Payment Activation Request message.

IPSL

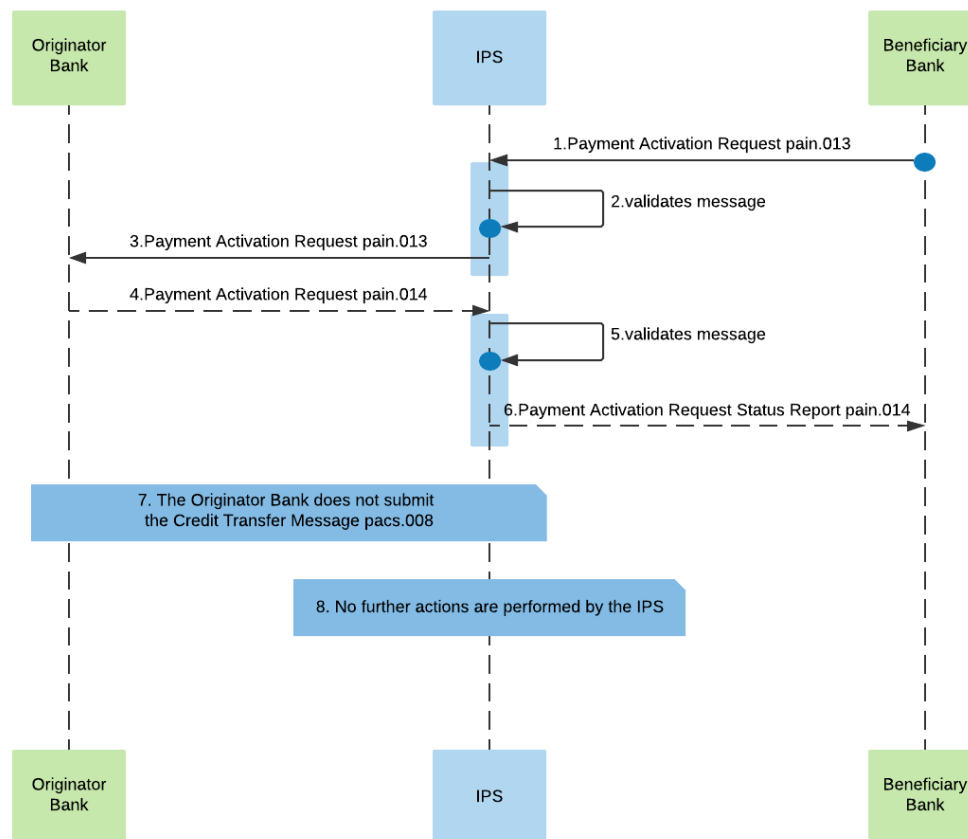
10.4 Payment Activation Request Status Report pain.014 message is invalid



1. The beneficiary Bank submits Payment Activation Request pain.013 to the IPS. The message contains data necessary for payment initiation: Originator Bank name, Beneficiary Bank name, Ultimate Debtor's name and account number (or phone number), Ultimate Creditor's name and account number (or phone number), payment details.
2. The IPS system successfully validates the payment initiation message.
3. The IPS forwards the Payment Activation Request message pain.013 to the Originator Bank the Ultimate Debtor (addressee of the request) needs to confirm his or her willingness to transfer money to the requestor.
4. If confirmation from the Ultimate Debtor is received, The Originator Bank submits Payment Activation Request Status (pain.014) with a positive result of the request to pay.
5. The IPS unsuccessfully validates the received message.
6. If Payment Activation Request Status Report pain.014 message is invalid, the IPS submits rejection message Payment Activation Request Status (pain.014) to the Originator Bank.
7. The IPS informs the Beneficiary Bank about the unsuccessful result of the initial request to pay within Payment Activation Request Status (pain.014).

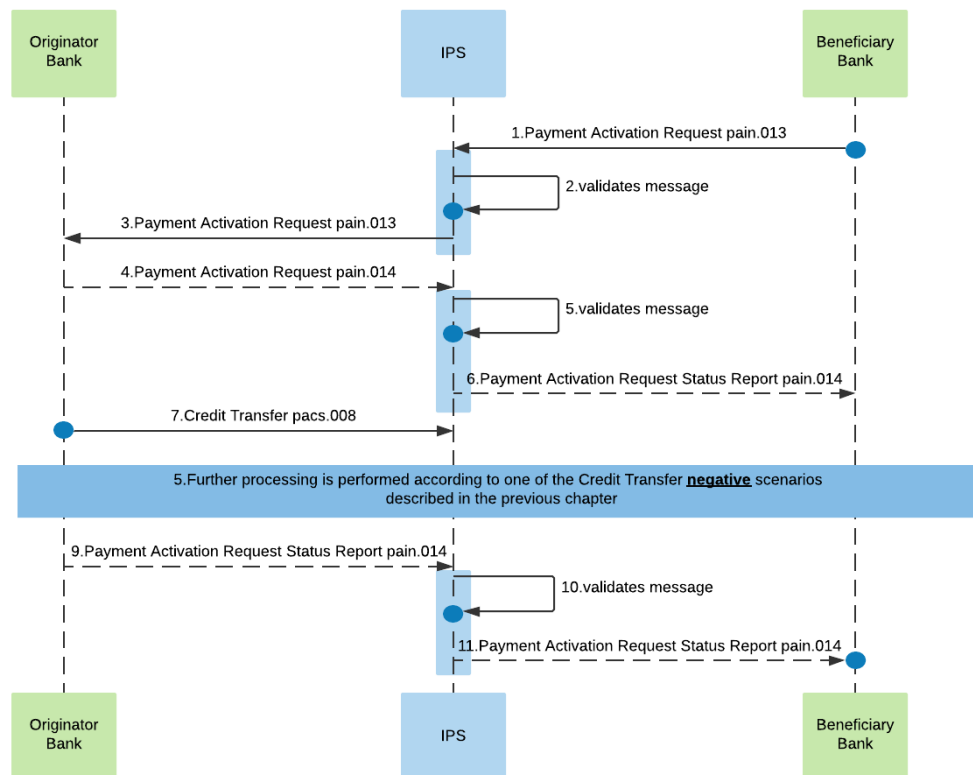
IPSL

10.5 The Credit Transfer is not initiated



1. The Beneficiary Bank initiates Payment Activation Request pain.013 to request the Originator Bank to execute credit transfer from the Ultimate Debtor in the favor of the Ultimate Creditor. The request message contains the Originator Bank name, Beneficiary Bank name, Ultimate Debtor's name and account number, Ultimate Creditor's name, and account number, payment details.
2. The IPS successfully validates the received message.
3. The IPS forwards Payment Activation Request pain.013 to The Originator Bank, where Ultimate Debtor (addressee of the request) needs to confirm his or her willingness to transfer money to the requestor. Only after positive confirmation is received, the Originator Bank is allowed to proceed with the Credit Transfer request submitting to the IPS.
4. If confirmation from the Ultimate Debtor is received, The Originator Bank submits Payment Activation Request Status (pain.014) with a positive result of the request to pay.
5. The IPS successfully validates the received message.
6. The IPS forwards the Payment Activation Request Status Report (pain.014) to the Beneficiary Bank.
7. The Originator Bank does not submit the Credit Transfer message pacs.008 to Tieto IPS.
8. No further actions are performed by the IPS.

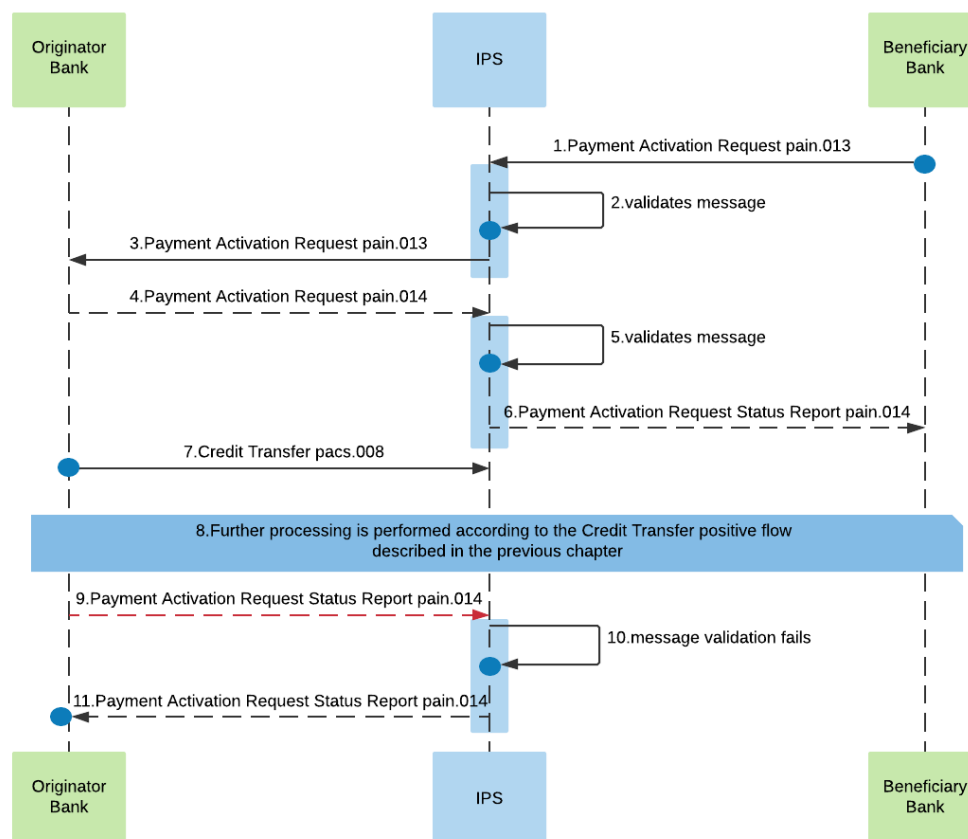
10.6 The Credit Transfer processing fails



1. The Beneficiary Bank initiates Payment Activation Request pain.013 to request the Originator Bank to execute credit transfer from the Ultimate Debtor in the favor of the Ultimate Creditor. The request message contains the Originator Bank name, Beneficiary Bank name, Ultimate Debtor's name and account number, Ultimate Creditor's name, and account number, payment details.
2. The IPS successfully validates the received message.
3. The IPS forwards Payment Activation Request pain.013 to The Originator Bank, where Ultimate Debtor (addressee of the request) needs to confirm his or her willingness to transfer money to the requestor. Only after positive confirmation is received, the Originator Bank is allowed to proceed with the Credit Transfer request submitting to the IPS.
4. If confirmation from the Ultimate Debtor is received, The Originator Bank submits Payment Activation Request Status (pain.014) with a positive result of the request to pay.
5. The IPS successfully validates the received message.
6. The IPS forwards the Payment Activation Request Status Report (pain.014) to the Beneficiary Bank.
7. The Originator Bank initiates Credit Transfer message pacs.008 to the IPS.
8. Further processing is performed according to one of the Credit Transfer to Account number **negative** scenarios described in the previous chapters.
9. The Originator Bank may initiate the new Payment Activation Status Report pain.014 to inform the Beneficiary Bank about the final status of the transaction chain caused by the initial Credit Payment Activation Request - in this case about the negative result of the transaction.
10. The IPS successfully validates the received message.

11. The IPS forwards Payment Activation Status Report pain.014 to the Beneficiary Bank.

10.7 The final Payment Activation Status Report submitted by the Originator Bank is invalid



1. The Beneficiary Bank initiates Payment Activation Request pain.013 to request the Originator Bank to execute credit transfer from the Ultimate Debtor in the favor of the Ultimate Creditor. The request message contains the Originator Bank name, Beneficiary Bank name, Ultimate Debtor's name and account number, Ultimate Creditor's name, and account number, payment details.
2. The IPS successfully validates the received message.
3. The IPS forwards Payment Activation Request pain.013 to The Originator Bank, where Ultimate Debtor (addressee of the request) needs to confirm his or her willingness to transfer money to the requestor. Only after positive confirmation is received, the Originator Bank is allowed to proceed with the Credit Transfer request submitting to the IPS.
4. If confirmation from the Ultimate Debtor is received, The Originator Bank submits Payment Activation Request Status (pain.014) with a positive result of the request to pay.
5. The IPS successfully validates the received message.
6. The IPS forwards the Payment Activation Request Status Report (pain.014) to the Beneficiary Bank.
7. The Originator Bank initiates Credit Transfer message pacs.008 to the IPS.

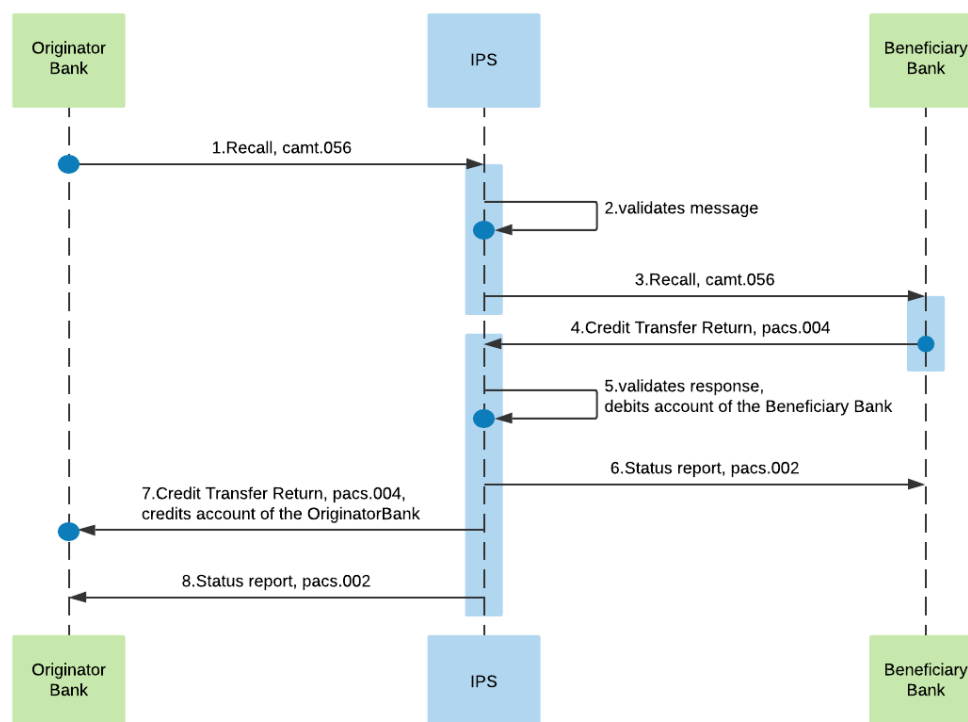
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8. Further processing is performed according to the Credit Transfer to Account number positive scenario described in the previous chapter.
9. The Originator Bank may initiate the new Payment Activation Status Report pain.014 to inform the Beneficiary Bank about the final status of the transaction chain caused by the initial Credit Payment Activation Request - in this case about the negative result of the transaction.
10. The IPS unsuccessfully validates the received message.
11. The IPS returns the negative Payment Activation Status Report pain.014 to the Originator Bank indicating that submitted message is incorrect. The Originator Bank may resend corrected response again.

11 RECALL

The IPS supports the possibility to disable Recall functionality pre-go-live and enable it after go-live based in the IPSL use case and need at that time. This will not result in a change request.

11.1 Payment cancellation accepted by the Beneficiary Bank, positive flow



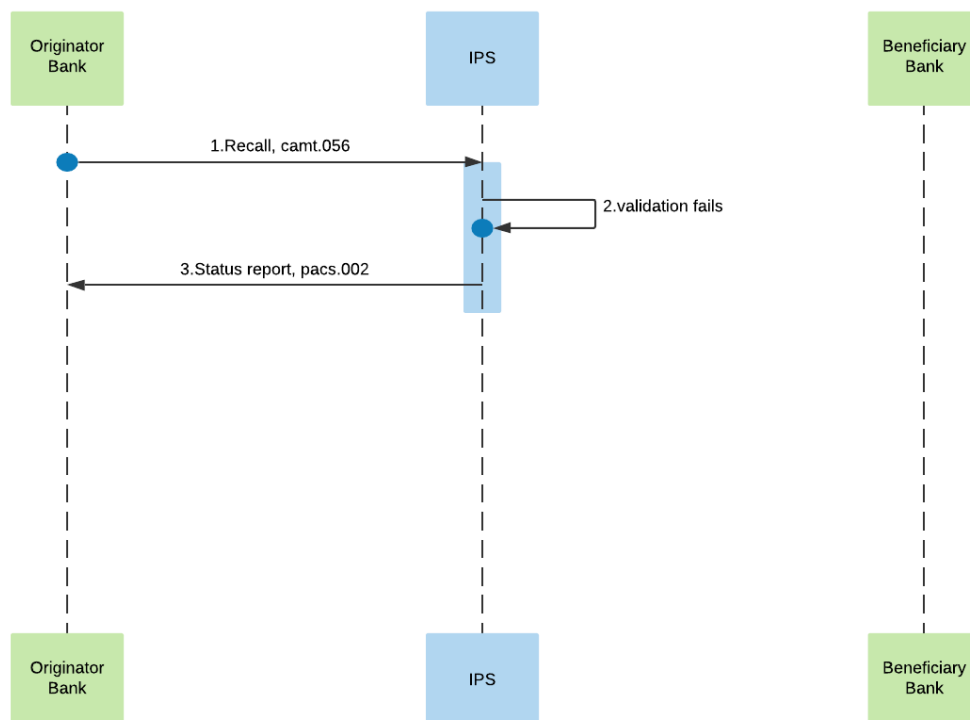
1. The Originator Bank has an opportunity to request cancelation of previously processed transaction. To cancel Credit Transfer, the Originator Bank has to submit Recall message, camt.056.
2. The IPS successfully validates the received message.
3. The IPS forwards Recall message, camt.056 to the Beneficiary Bank.
4. If after internal investigation the Beneficiary Bank decides to accept Credit Transfer cancelation request, the Beneficiary Bank submits Credit Transfer Return message, pacs.004 to the IPS.

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5. The IPS validates the Credit Transfer Return message, pacs.004, and debits the liquidity position of the Beneficiary Bank for the amount of the return transaction.
6. The IPS returns the Credit Transfer Status Report pacs.002 message to the Beneficiary Bank to inform it about the status of processing of Credit Transfer Return message, pacs.004.
7. The IPS credits the liquidity position of the Originator Bank forwards the Credit Transfer Return message, pacs.004 to the Originator Bank.
8. The IPS submits the final Credit Transfer Status Report message pacs.002 to the Originator Bank as final confirmation on pacs.004 processing result.

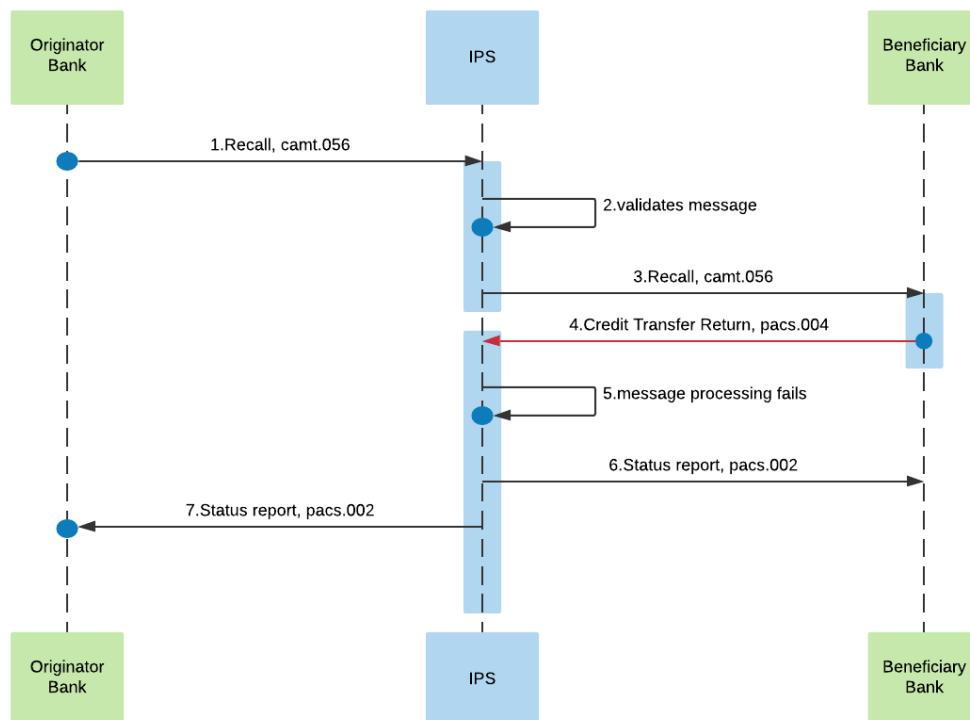
11.2 Payment cancellation accepted by the Beneficiary Bank, negative scenarios

11.2.1 Recall message validations by IPS are failed (message is invalid)



1. The Originator Bank submits Recall message, camt.056 to the IPS.
2. The IPS unsuccessfully validates the received message.
3. The IPS returns the Credit Transfer Status Report message pacs.002 to the Originator Bank to inform the participant about the Recall message processing fail.

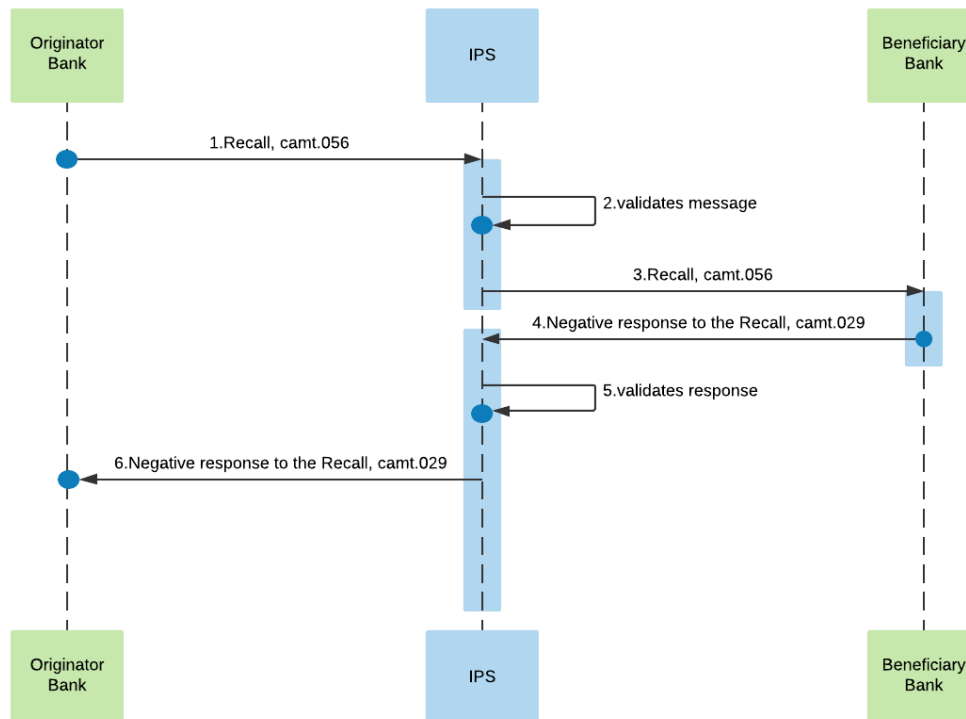
11.2.2 Credit Transfer Return message validations by IPS are failed (message is invalid, not possible to debit account of the Beneficiary Bank)



1. The Originator Bank submits Recall message, camt.056 to the IPS
2. The IPS successfully validates the received message.
3. The IPS forwards Recall message, camt.056 to the Beneficiary Bank.
4. If after internal investigation the Beneficiary Bank decides to accept Credit Transfer cancellation request, the Beneficiary Bank submits Credit Transfer Return message, pacs.004.
5. The IPS unsuccessfully validates the Credit Transfer Return message, pacs.004, or it is not possible to debit the liquidity position of the Beneficiary Bank for the amount of the return transaction.
6. The IPS returns the Credit Transfer Status Report message pacs.002 to the Beneficiary Bank to inform the participant about the Credit Transfer Return message processing fail.
7. The IPS returns the Credit Transfer Status Report message pacs.002 to the Originator Bank to inform the participant about the Recall message processing fail.

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11.3 Payment cancellation rejected by the Beneficiary Bank, positive flow



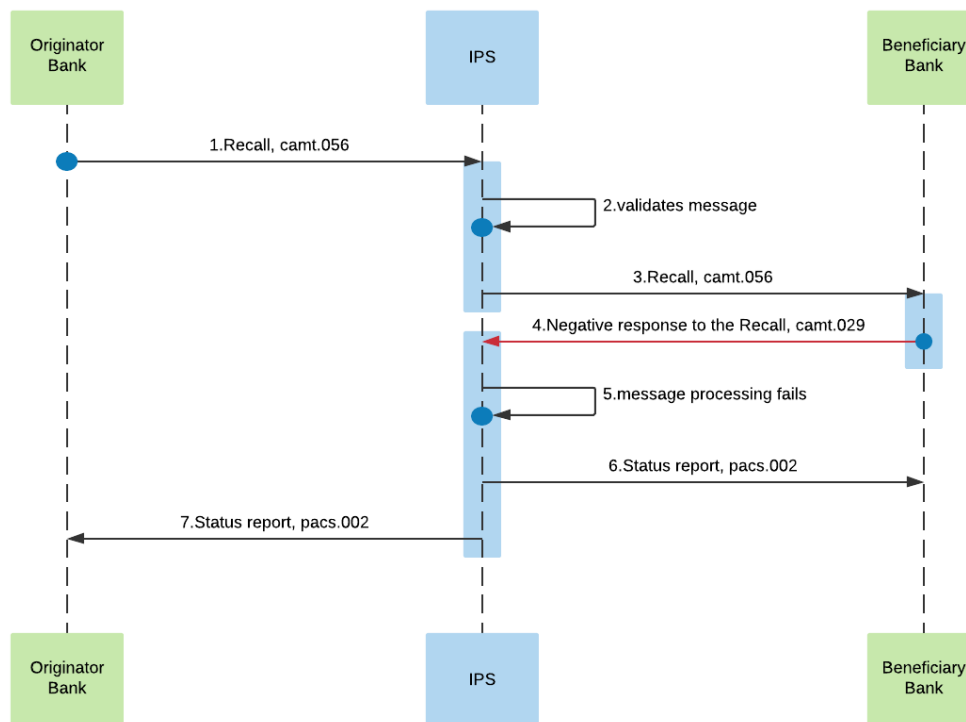
1. The Originator Bank has an opportunity to request cancellation of previously processed transaction. To cancel Credit Transfer, the Originator Bank has to submit Recall message, camt.056.
2. The IPS successfully validates the received message.
3. The IPS forwards Recall message, camt.056 to the Beneficiary Bank.
4. If after internal investigation the Beneficiary Bank decides not to accept Credit Transfer cancellation request, the Beneficiary Bank submits Negative response to the Recall, camt.029 to IPS.
5. The IPS validates the Negative response to the Recall, camt.029.
6. The IPS forwards Negative response to the Recall, camt.029 to the Originator Bank.

11.4 Payment cancellation rejected by the Beneficiary Bank, negative scenarios

11.4.1 Recall message validations by IPS are failed (message is invalid)

please see description provided in the chapter 11.2.1.

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11.4.2 Negative response to the Recall message validations by IPS are failed (message is invalid)

1. The Originator Bank has an opportunity to request cancelation of previously processed transaction. To cancel Credit Transfer, the Originator Bank has to submit Recall message, camt.056.
2. The IPS successfully validates the received message.
3. The IPS forwards Recall message, camt.056 to the Beneficiary Bank.
4. If after internal investigation the Beneficiary Bank decides not to accept Credit Transfer cancelation request, the Beneficiary Bank submits Negative response to the Recall, camt.029 to IPS.
5. The IPS unsuccessfully validates the Negative response to the Recall, camt.029.
6. The IPS returns the Credit Transfer Status Report message pacs.002 to the Beneficiary Bank to inform the participant about the Negative response to the Recall message processing fail.
7. The IPS returns the Credit Transfer Status Report message pacs.002 to the Originator Bank to inform the participant about the Recall message processing fail.

12 Investigation, positive scenario

The participants may initiate investigation procedure when it is necessary to get information about the final status of the transaction.

IPSL

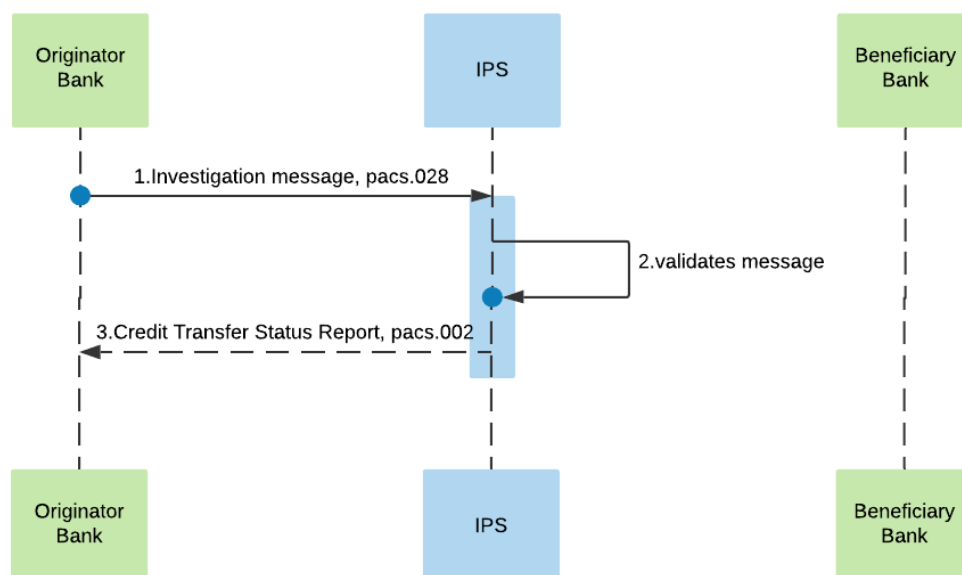
If there no final confirmation (Credit Transfer Status Report message pacs.002) received from the IPS as result of Credit Transfer processing, the Originator Bank may request investigation before to finalize the debit transaction with its customer, Ultimate Debtor.

If there no final confirmation (Credit Transfer Status Report message pacs.002) received from the IPS as result of Credit Transfer processing, the Beneficiary Bank may request investigation before to finalize the credit transaction with its customer, Ultimate Creditor.

As well, if there is no response received as result of Recall message processing, the Originator Bank may initiate investigation procedure, to request the information about the status of recall transaction.

The investigation procedure allowed to be initiated only after the corresponding Credit Transfer processing timeout expired, otherwise investigation message will be rejected.

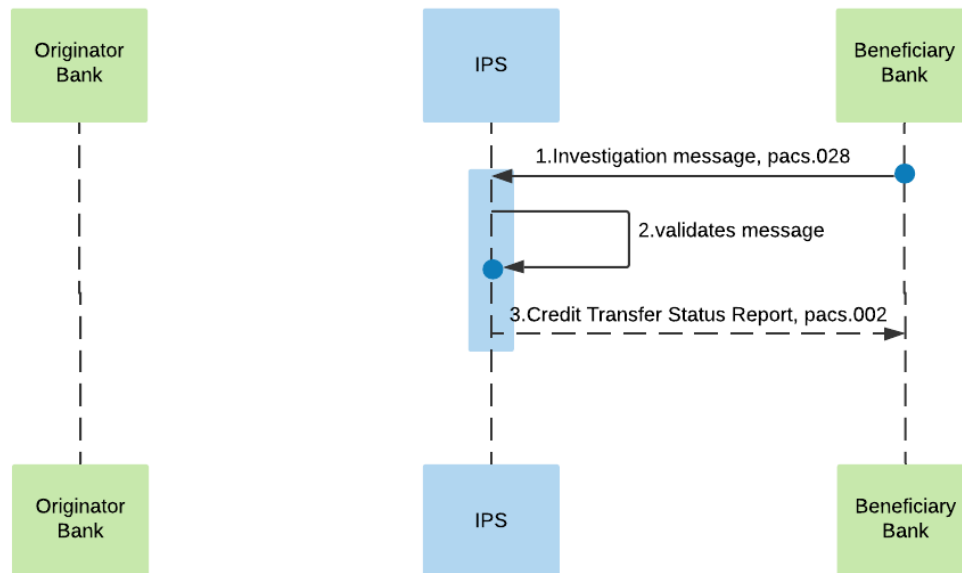
12.1 Credit Transfer Investigation message initiated by the Originator Bank



1. The Originator Bank initiates Investigation message pacs.028 to the IPS.
2. The IPS validates Investigation message pacs.028, and finds the initial Credit Transfer transaction.
3. The IPS returns to the Originator Bank Credit Transfer Status Report message pacs.002 with the actual status of the initial Credit Transfer transaction. Possible answers are:
 - a. The Beneficiary Bank rejected the Credit Transfer request;
 - b. The Beneficiary Bank accepted the Credit Transfer request and the transaction had been settled;
 - c. Transaction is closed due to timeout.

IPSL

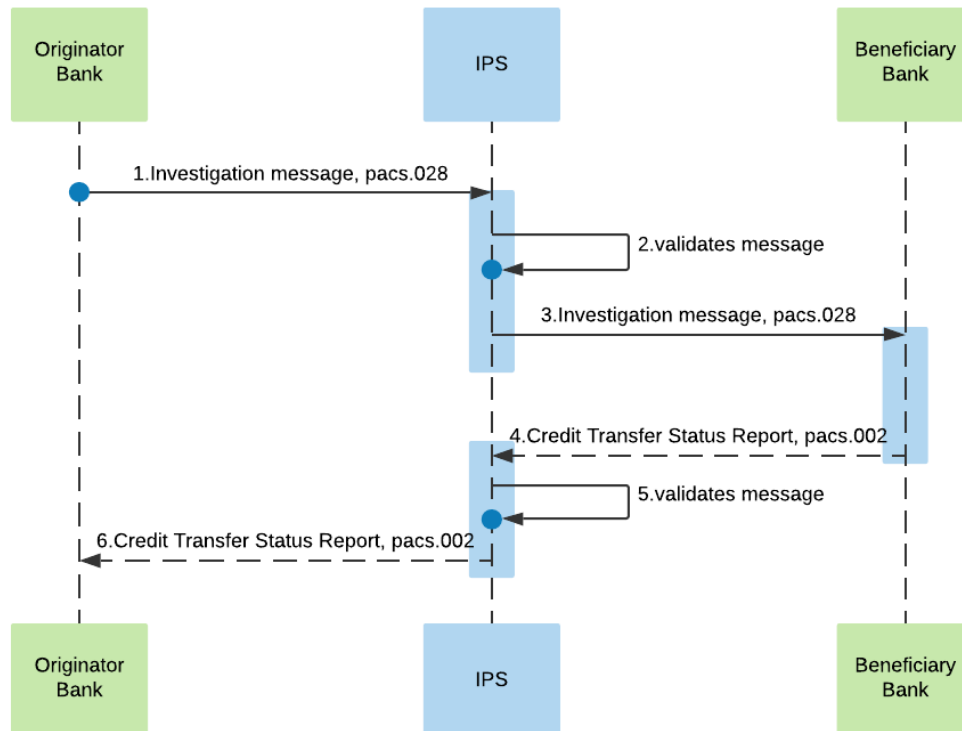
12.2 Credit Transfer Investigation message initiated by the Beneficiary Bank



1. The Beneficiary Bank initiates Investigation message pacs.028 to the IPS.
2. The IPS validates Investigation message pacs.028, and finds the initial Credit Transfer transaction.
3. The IPS returns to the Beneficiary Bank Credit Transfer Status Report message pacs.002 with the actual status of the initial Credit Transfer transaction. Possible answers are:
 - a. The Beneficiary Bank rejected the Credit Transfer request;
 - b. The Beneficiary Bank accepted the Credit Transfer request and the transaction had been settled;
 - c. Transaction is closed due to timeout.

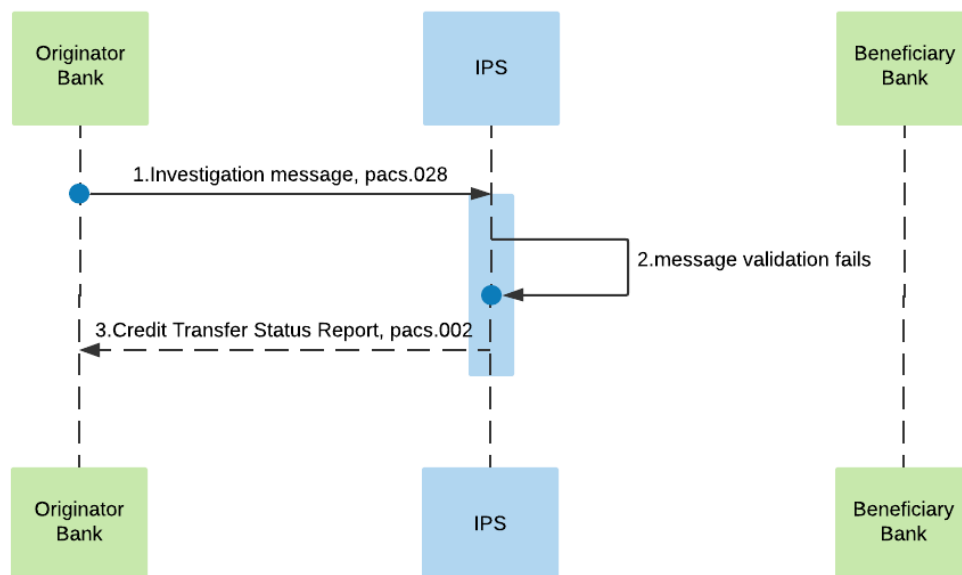
IPSL

12.3 Investigation message on Recall transaction



1. The Originator Bank initiates Investigation message pacs.028 to the IPS.
2. The IPS validates the Investigation message pacs.028.
3. The IPS forwards the Investigation message pacs.028 to the Beneficiary Bank.
4. The Beneficiary Bank finds the initial Recall request and returns to IPS Credit Transfer Status Report message pacs.002 with the actual status of the initial Recall transaction. Possible answers are:
 - a. The Beneficiary Bank rejected the Recall request;
 - b. The Beneficiary Bank accepted the Recall request and the transaction had been settled.
5. The IPS validates the Credit Transfer Status Report message pacs.002.
6. The IPS forwards the Credit Transfer Status Report message pacs.002 indicating the final status of the Recall transaction to the Originator Bank.

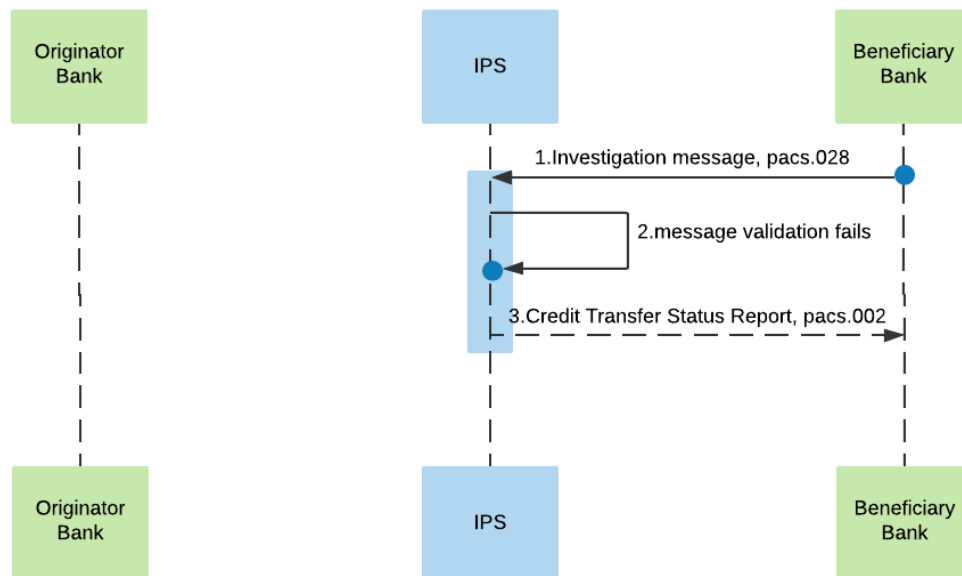
IPSL

13 Investigation, negative scenarios**13.1 Credit Transfer Investigation message initiated by the Originator Bank is rejected by IPS**

1. The Originator Bank initiates Investigation message pacs.028 to the IPS.
2. The IPS unsuccessfully validates the Investigation message pacs.028. Potential reasons for the Investigation message decline are:
 - a. Invalid message pacs.028;
 - b. Investigation message is submitted before the corresponding Credit Transfer processing timeout expired;
 - c. Not possible to find the initial Credit Transfer transaction.
3. The IPS returns to the Originator Bank Credit Transfer Status Report message pacs.002 indicating the reason of message rejection.

IPSL

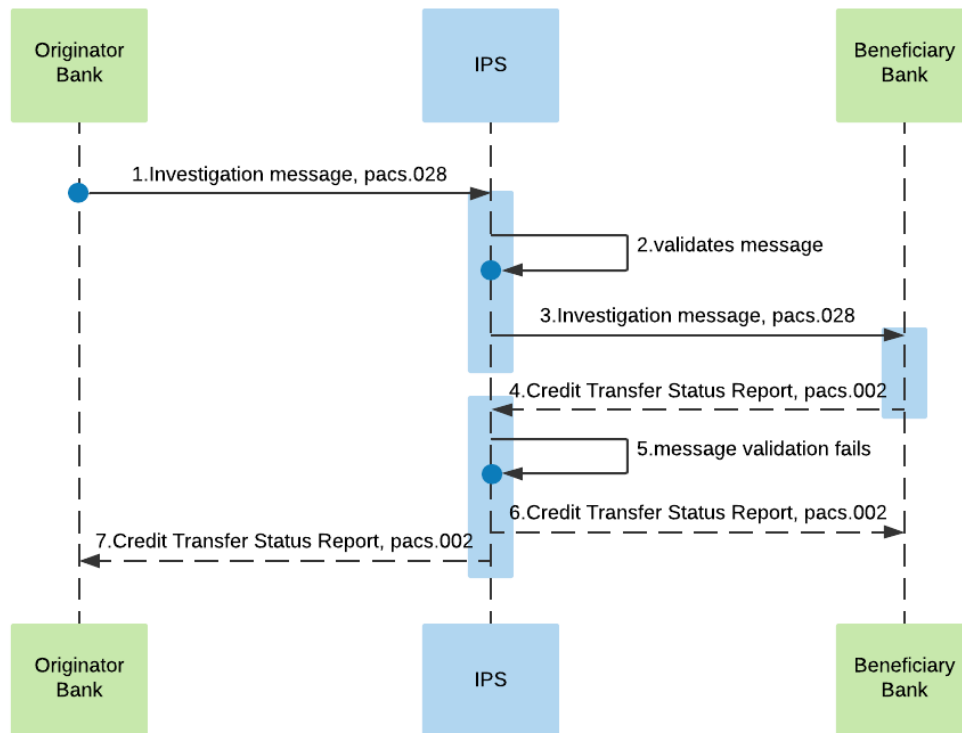
13.2 Credit Transfer Investigation message initiated by the Beneficiary Bank is rejected by IPS



1. The Beneficiary Bank initiates Investigation message pacs.028 to the IPS.
2. The IPS unsuccessfully validates the Investigation message pacs.028. Potential reasons for the Investigation message decline are:
 - a. Invalid message pacs.028;
 - b. Investigation message is submitted before the corresponding Credit Transfer processing timeout expired;
 - c. Not possible to find the initial Credit Transfer transaction.
3. The IPS returns to the Beneficiary Bank Credit Transfer Status Report message pacs.002 indicating the reason of message rejection.

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13.3 Invalid response to the Recall Investigation message is submitted by the Beneficiary Bank



1. The Originator Bank initiates Investigation message pacs.028 to the IPS.
2. The IPS successfully validates the Investigation message pacs.028.
3. The IPS forwards the Investigation message pacs.028 to the Beneficiary Bank.
4. The Beneficiary Bank finds the initial Recall request and returns to IPS Credit Transfer Status Report message pacs.002 with the actual status of the initial Recall transaction.
5. The IPS unsuccessfully validates the Credit Transfer Status Report message pacs.002.
6. The IPS returns to the Beneficiary Bank Credit Transfer Status Report message pacs.002 indicating the reason of message rejection (for example, invalid message).
7. The IPS generates the Credit Transfer Status Report message pacs.002 to the Originator Bank indicating that the Recall Investigation message processing failed. The Originator Bank may re-submit the Investigation message one more time.

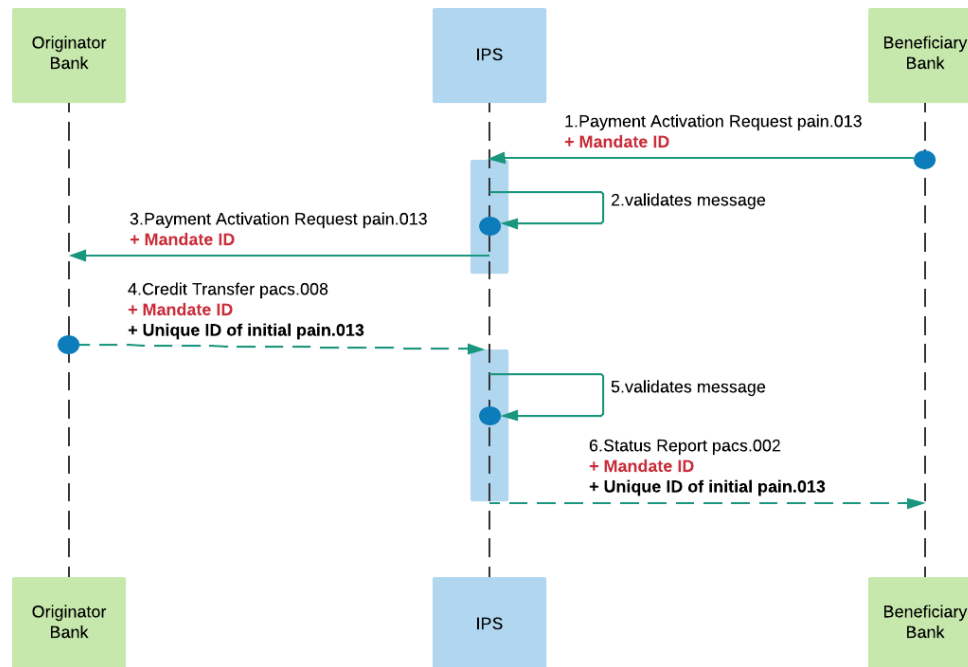
14 Change Request - Request to Pay with Mandate ID, IPLIPSLIPS-10

The IPS solution is extended to support additional customized flow Request to Pay with Mandate ID (pain.013/pain.014).

1. Proprietary field "Mandate ID" will be added to the following types of messages:
 - Payment Activation Request message (pain.013)
 - Payment Activation Request Status Report (pain.014)
 - Credit Transfer (pacs.008)
 - Credit Transfer Status Report message (pacs.002).
2. Request to Pay with Mandate ID processing flow is the proprietary flow for the IPSL use only. This flow is based on the IPSL specific requirements. Request to Pay with Mandate ID processing flow must be considered as a customized solution of the standard Credit Transfer and Request to Pay processing flows.
3. New Request to Pay with Mandate ID processing flow assumes separate API endpoints for message initiation and response receiving.

Solution limitation:

1. The IPS does not validate Mandate ID field, and does not provide any business logic based on Mandate ID usage. The responsibility of the IPS is to submit Mandate ID in the corresponding field, transfer it to the counterparty and store it within the corresponding messages as historical data.
2. No separate reporting for Request to Pay with Mandate ID transactions is supported. Mandate ID as the separate field will not be added to reports that are already included in IPS solution. Credit Transfer processed in the result of Request to Pay with Mandate ID flow processing will be available in the reports where standard Credit Transfers (pacs.008) are reflected.
3. No separate limits for Request to Pay with Mandate ID transactions are supported. These transactions will be controlled under the total limits defined for the Request to Pay (pain.013) transactions.
4. The IPS does not perform any validations regarding Mandate ID and Mandate ID presence in the message. Mandate related error codes MD01 No Mandate and MD02 MissingMandatoryInformationInMandate might be supported in pain.014 messages initiated by the Originator Bank (receiving party) received as a negative response to pain.013 with Mandate ID. Error codes MD01 and MD02 received in the response message (pain.014) sent by the Originator Bank (receiving party) will be just forwarded to the Beneficiary Bank, without any validations and verifications.

Positive flow for the Request to Pay with Mandate ID transaction processing:

The Payment Activation Request message pain.013 is used as Request to Pay with Mandate ID message. The Request to Pay with Mandate ID message is sent by the Beneficiary Bank (the initiating party might be the Indirect Participant as well) to request an account to be debited at the receiving bank (the Originator Bank). The message contains Mandate related information, i.e. extracts from a mandate, such as Mandate ID in which is the authorization and expression of consent given by the Ultimate Debtor to the Ultimate Creditor to allow such debit on the debtor's account. The Mandate ID is generated during the pre-authorization procedure and is out of the scope of the IPS services.

Request to Pay with Mandate Flow:

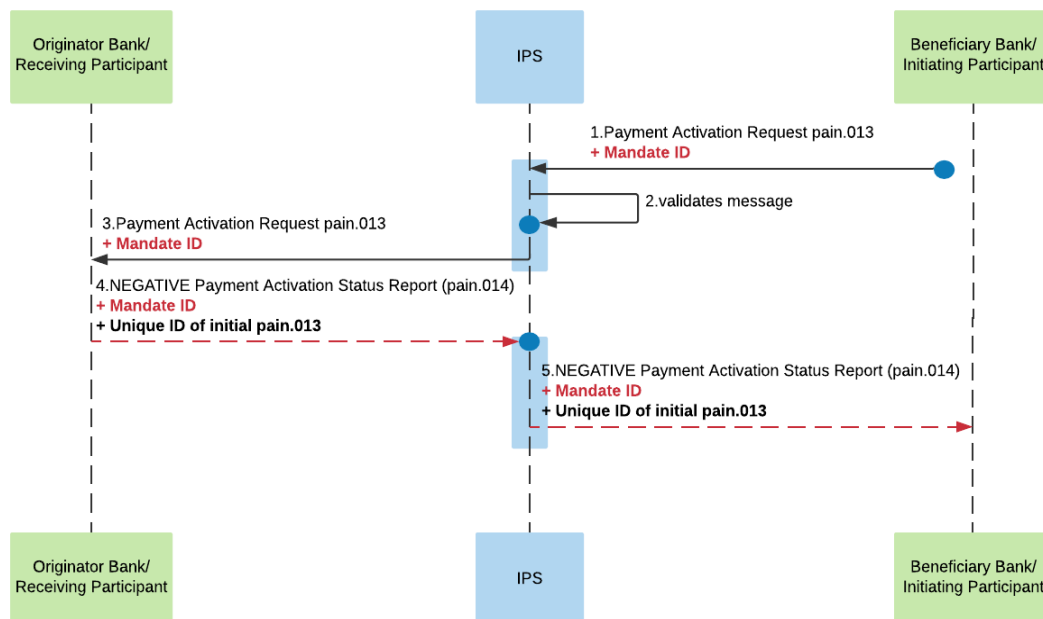
1. The Beneficiary Bank (Initiating Participant) initiates Payment Activation Request pain.013 to request the Originator Bank (Receiving Participant) as the Request to Pay with Mandate ID message. The request message contains the Originator Bank name, Beneficiary Bank name, Ultimate Debtor's name and account number, Ultimate Creditor's name, and account number, payment details. As well the request message contains the Mandate ID (Mandate Code).
2. The IPS successfully validates the received message. The Mandate ID validations are not performed by the IPS.
3. The IPS forwards Payment Activation Request pain.013 to the Originator Bank (Receiving Participant), where the Originator Bank (Receiving Participant) performs necessary validation on the Mandate Identification (Mandate Code).
4. The Originator Bank (Receiving Participant) confirms debit on the Ultimate Debtor's account and responds with the Credit Transfer message pacs.008. The response message contains the Mandate ID and reference ID of the initial message pain.013.
5. The IPS successfully validates the received message. The IPS checks statuses of the Originator Bank (Receiving Participant) and Beneficiary Bank (Initiating Participant), checks existing limits for the participants, and checks the net position of the Originator Bank (Receiving Participant).

The IPS changes net positions on the participants by moving transaction amount (including fee, if applicable) from the Originator Bank's IPS account to Beneficiary Bank's IPS account.

- Optionally the IPS submits final Credit Transfer Status Report message pacs.002 to the Beneficiary Bank (Initiating Participant) to inform it about the positive result of initial Request to Pay with Mandate ID processing.

Negative flows for the Request to Pay with Mandate ID transaction processing:

Request to Pay with Mandate ID is rejected by the Originator Bank

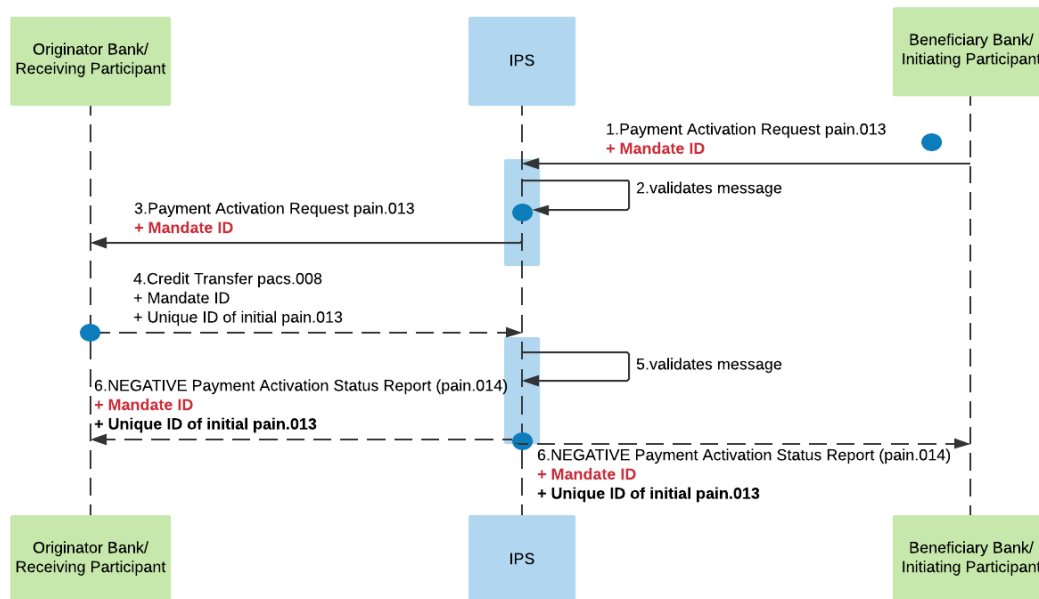


- The Beneficiary Bank (Initiating Participant) initiates Payment Activation Request pain.013 to request the Originator Bank (Receiving Participant) as the Request to Pay with Mandate ID message. The request message contains the Originator Bank name, Beneficiary Bank name, Ultimate Debtor's name and account number, Ultimate Creditor's name, and account number, payment details. As well the request message contains the Mandate ID.
- The IPS successfully validates the received message. The Mandate ID validations are not performed by the IPS.
- The IPS forwards Payment Activation Request pain.013 to The Originator Bank (Receiving Participant), where the Originator Bank (Receiving Participant) performs necessary validation on the Mandate ID.
- The Originator Bank (Receiving Participant) rejects debit on the Ultimate Debtor's account and responds with negative Payment Activation Status Report message pain.014. The response message contains the Mandate Identification and reference ID of the initial message pain.013.
- The IPS forwards negative Payment Activation Status Report message pain.014 to the Beneficiary Bank (Initiating Participant) to inform it about the negative result of initial Request to Pay with Mandate ID processing.

As well if there is no response received from the Originator Bank within the pre-defined time limit for Request to Pay with Mandate ID processing flow, the IPS generates negative Payment

Activation Status Report message pain.014 to the Beneficiary Bank (Initiating Participant) to inform it about the negative result of initial request due to timeout.

IPS failed to make changes of net position on the participant's accounts



1. The Beneficiary Bank (Initiating Participant) initiates Payment Activation Request pain.013 to request the Originator Bank (Receiving Participant) as the Request to Pay with Mandate ID message. The request message contains the Originator Bank name, Beneficiary Bank name, Ultimate Debtor's name and account number, Ultimate Creditor's name, and account number, payment details. As well the request message contains the Mandate ID.
2. The IPS successfully validates the received message. The Mandate ID validations are not performed by the IPS.
3. The IPS forwards Payment Activation Request pain.013 to The Originator Bank (Receiving Participant), where the Originator Bank (Receiving Participant) performs necessary validation on Mandate ID.
4. The Originator Bank (Receiving Participant) confirms debit on the Ultimate Debtor's account and responds with the Credit Transfer message pacs.008. The response message contains the Mandate ID and reference ID of the initial message pain.013.
5. The IPS unsuccessfully validates the received message or the Originator Bank's net position is not enough to perform online clearing transaction.
6. The IPS generates negative Payment Activation Status Report message pain.014. to the Originator Bank (Receiving Participant) and to the Beneficiary Bank (Initiating Participant). The response message contains the Mandate Identification, reference ID of the initial message pain.013 and corresponding response code.